Chapter 10

Mushrooms: A Source of Exciting Bioactive Compounds

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INTRODUCTION

It is known that the usage of mushrooms as therapeutic agents is not new. In the old times, they are used by the local people as food as well as medicines, particularly in the Asian countries such as, China, Japan, Korea as well as in some part of Africa. Nowadays, mushrooms are used in medicine, pharmacy, food, and fermentation fields, as well. Particularly, mushrooms are mostly used as therapeutic agents besides their nutritional properties. Therefore, mushrooms have been studied by researchers and, isolating and elucidating the structure of the biologically active compounds has been the key area studied by scientists. Mostly, lectins, polysaccharides (β-glucans), polysaccharide-peptides, polysaccharide-protein complexes, lanostanoids, other terpenoids, alkaloids, sterols, and phenolic-structured compounds, indicating anticancer, antioxidant, antitumor, anti-inflammatory, antifungal, antibacterial, antiviral, anti-immunomodulatory activities have been isolated from the mushroom species [1–6]. Mushrooms are

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especially used as immunomodulator and antitumor agents. Generally, mushrooms are thought that they are natural foods and are produced or naturally grown all over the world [6].

Since mushrooms are rich in vitamins, proteins, and minerals, they have been considered as valuable healthy foods. The studies on edible mushrooms show that they are low in calories and fats; [7] yet of the fats present they can be considered rich in polyunsaturated fatty acids. In addition to nutritional properties, mushrooms are preferred in most countries especially due to the fact that their tastes and flavor [8–9]. Moreover, various biological activities such as antifungal, antibacterial, antioxidant [10], anti-immunomodulatory, antiviral [1], antitumor [2,6], anti-inflammatory [5,11], antiaromatase [12], anticholesterole [13], cytotoxic [6], anti-coagulant, and antihypoglycemic [14–15] activities of edible mushrooms were investigated. Almost 660 species of higher basidiomycetes were tested which possessed antitumor activity [16].

Among the causes of deaths, cancer-related deaths are ranked second after the cardiovascular diseases. The most common cancer in women is the breast cancer, and the second most common in men is the lung cancer disease. The incidence of breast and lung cancer, among all cancers is around 17–20%, and 12–15%, respectively. According to a joint report of the World Health Organization (WHO) and International Agency for Research on Cancer (IARC), the incidence of breast cancer in the world is increasing by 0.5% per year, and each year breast cancer develops in 1 million women in the world and 370,000 women die from the disease. As for men, according to WHO report, 1.3 million men die around the world because of the lung cancer. These facts mean that approximately 4–5 million people died due to all cancer types [17]. Most cancer treatments fail, and even to provide a cure most of cancer treatment drugs damage the healthy cells. The literature survey shows that the mushrooms have been used particularly for their anticancer and immunomodulation activities [1,6].

Medicinal mushrooms are sources of polysaccharides that prevent the cancer and increase the strength of the immune system (as immunomodulatory) properties. Therefore, mushrooms serve as an important source of drugs for modern medicine. As an example, the anticancer drug obtained from *Ganoderma lucidum* and produced in Japan by the name of Krestin, which is 25% of the world's total cancer drug marketing [18]. Thus, *Ganoderma* species attract the attention. In addition, a number of mushroom species having polysaccharides with high molecular weight have been subjected through clinical trials such as phase I, phase II, and phase III. As another example, lentinan a polysaccharide isolated from *Lentinus edodes* has been used in a great number of cancer patients in clinical trials [19].

Isolation of biologically active compounds from various mushroom species has been increasing due to the wide range of medical activities of mushrooms. Mushrooms including their mycelium and other parts can provide many compounds with various biological activities.

In this chapter, reports on mushroom species, which were studied for their chemistry together with their biological activities in the last two decades were

evaluated. Particularly, the studies comprise the last decade. In the subtitles polysaccharides, triterpenoids, steroids, phenolic compounds, and alkaloids isolated from the mushroom species are discussed together with their biological activities. Briefly, 24 items of polysaccharides including β-glucans and other complexes from 13 mushroom species; 259 terpenoid compounds including 7 monoterpenes, 19 sesquiterpenes, 54 diterpenes, and 179 triterpenes from 29 mushroom species; 55 steroid compounds from 11 mushroom species; 41 phenolic compounds from 13 mushroom species; and 42 alkaloid compounds from 13 species have been evaluated. In addition, antibacterial, anticandidal, antileishmanicidal, anticancer, antitumor, antimycobacterial, cytotoxicity, antimalarial, antitubercular, anticholinesterase, antioxidant, anticompliment, anti-invasive, antiviral, and anti-inflammatory activities of some of these isolated compounds were also included.

ISOLATED BIOACTIVE COMPOUNDS

The Polysaccharides

Polymers of sugars joined to each other by α or β glycosidic linkages called polysaccharides. Polysaccharides including heteroglucans are macromolecules, which have high potential for structural variability. Glucans have been isolated from various mushrooms and mycelia sources to date, and they are interesting due to having potent therapeutic agents [20]. They have also the highest capacity of biological activities [20]. In general, there is a main chain, which is either β (1–3), β (1–4), or mixed β (1–3), β (1–4) with β (1–6) side chains. Many studies revealed that some of these polysaccharides isolated from mushrooms are responsible to prevent mostly cancer diseases [19,21]. In fact, most of the polysaccharides isolated from mushroom species had potent antitumor activity against some cancerous cell lines such as sarcoma 180 and a host of other tumor cell lines [20]. Hetero-β-D-glucans show anticancer activity; however, α-D-glucans are unable to show anticancer activity [20]. In addition, they also show immunomodulatory activity [20–31]. The bioactive polysaccharides with their properties and activities were given in Table 1, and the structures of **1-16** in Figure 1.

The hot aqueous extract of the edible mushroom *Tricholoma crassum* (Berk.) Sacc. contained glucan (**1**), which is soluble in water and its average molecular weight is calculated 1.74×10^5 Da. The macrophage activity by nitric oxide (NO), and splenocyte and thymocyte proliferation activity, and lipid peroxidation inhibition activity by TBARS (thiobarbutiric acid reactive substances) were methods used to determine the bioactivity of this glucan (**1**). It was reported that *in vitro* macrophage activation by NO production of 1 was found to be in a dose-dependent style. In addition, **1** demonstrated effective splenocyte and thymocyte immunostimulation in the cell culture of mouse. Compound **1** well inhibited the lipid peroxidation, either. According

TABLE 1 Bioactive Polysaccharides with Their Properties and Activities

Mushroom Species	Type	Solubility	Properties	Actions	References
Tricholoma crassum	Glucan (**1**)	Water soluble	Average molecular weight ~1.74×10^5 Da, specific rotation $[\alpha]_D^{30.7} = -9.6$ (c 0.104, H_2O)	*In vitro* macrophage activation by NO production in a dose-dependent style Immunostimulatory Lipid peroxidation inhibitory activity	[32]
Pleurotus ostreatus	Intracellular glucan (**2**)	Water soluble	Molecular weight ~2.7×10^6 Da, specific rotation $[\alpha] = \pm 25$ (c 0.1, H_2O)	Immune cell activation Anti-tumor potential	[21]
Termitomyces robustus	Glucan (PS-I and PS-II) (**3**–**4**)	Water soluble and water insoluble	PS-I: Molecular weight ~1.82×10^5 Da, specific rotation $[\alpha]_D^{25} = -31.6$ (c 0.6, H_2O)	Macrophage, splenocyte, thymocyte activations	[33]
Somatic hybrid of *Pleurotus florida* and *Calocybe indica* var.	β-glucan (**5**)	Water insoluble	Molecular weight ~169 kDa The presence of 1,5-di-*O*-acetyl-2,3,4, 6-tetra-*O*-methyl-glucitol (21.41%), 1,3,5-tri-*O*-acetyl-2,4,6-tri-*O*-methyl-glucitol (61.30%), and 1,3,5,6-tetra-*O*-acetyl-2,4-di-*O*-methyl-glucitol (17.29%) in a ratio of nearly 1:3:1, respectively.	Macrophage, splenocyte, thymocyte activations, immunostimulatory	[35]

TABLE 1 Bioactive Polysaccharides with Their Properties and Activities—Continued

Mushroom Species	Type	Solubility	Properties	Actions	References
P. florida	Glucan (PS-I and PS-II) (**6**–**7**)	Water soluble	Molecular weight ~2.02×10^5 Da, ~1.76×10^5 Da, a glucan, (1→6)-*β*-D-linked moiety), specific rotation of PS-I $[\alpha]_D = +17.07$ (c 0.098, H_2O)	Macrophage, splenocyte, thymocyte activations	[14]
Somatic hybrid of *P. florida* and *Volvariella Volvacea*	β-glucan (**8**)	Water Soluble	Molecular weight ~1.85×10^5 Da (1→6)-*β*-D-glucan), specific rotation $[\alpha]_D^{25} = -31.2$	Macrophage, splenocyte, thymocyte activations, immunoenhancing	[36]
Termitomyces striatus	Heteropolysaccharide (**9**)	Water soluble	Molecular weight ~7×10^4, specific rotation $[\alpha]_D^{25} = +1.85$ (c 0.86, H_2O)	Splenocyte and macrophage activations	[37]
P. florida, cultivar Assam Florida	Glucan (**10**)	Water soluble	Molecular weight ~1.3×10^5 Da, specific rotation $[\alpha]_D^{25} = +56.0$ (c 0.80, H_2O)	Macrophage, splenocyte, thymocyte activations	[38]
Sarcodon aspratus	Polysaccharide (**11**)	Water soluble	Molecular weight 6.7×10^5 Da, a liner glucan with a backbone structure of (1→6)-linked-*α*-D-glucopyranosyl residues	Immunomodulator to protect from tumors and pathogens	[15]
Lentinus squarrosulus	Glucan (**12**)	Water soluble	Molecular weight 1.98×10^5 Da, specific rotation $[\alpha]_D^{25.5} = -21$ (c 0.1, H_2O) (1→3, 6)- linked, (1→3)-linked, (1→6)-linked, and terminal *β*-D-glucopyranosyl in a proportion of approximately 1:2:1:1.	Macrophage, splenocyte, thymocyte activations	[39]

Continued

TABLE 1 Bioactive Polysaccharides with Their Properties and Activities—Continued

Mushroom Species	Type	Solubility	Properties	Actions	References
Auricularia polytricha	Polysaccharide (AAPS-1, AAPS-2, AAPS-3) (**13–15**)	Water soluble	Molecular weight 162, 259 and 483 kDa,	Growth inhibition of transplanted S180 sarcoma in mice	[19]
Amorphophallus campanulatus	Heteropolysaccharide (**16**)	Water soluble	Molecular weight ~1.8×10^5Da, Composed of D-galactose, D-glucose, 4-*O*-acyl-D-methyl galacturonate, and L-arabinose in a molar ratio 2:1:1:1.	Splenocyte activation	[41]
Pleurotus ostreatus and *Pleurotus eryngii*	β-glucan (L1, L2, S) (**17–19**)	Water, and alkali soluble and insoluble	Molecular weight of water extract (2200 and 12 kDa), Molecular weight of alkali extract (2300 and 20.3 kDa) Branched 1,3-1,6-*β*-D glucan	Synbiotic construction with selected probiotic strains	[42]
Ganoderma tsugae	Polysaccharide (GTM1-GTM4) (**19–22**)	Water soluble	GTM1-GTM2: heteropolysaccharide–protein complexes, Molecular weight ~62.8×10^4 Da and ~81.8×10^4Da GTM3-GTM4: (1/3)-*β*-D-glucans and (1/4)-*α*-D-glucans, Molecular weight ~465×10^4 Da and ~468×10^4 Da GTM5-GTM6: (1/6)-Branched (1/3)-*β*-D-glucan,	Antitumor activity	[43]
G. tsugae	Polysaccharide (GTM5-GTM6) (**23–24**)	Water soluble	(1/6)-branched (1/3)-*β*-D-glucan, Molecular weight ~176×10^4 Da and ~161×10^4 Da	–	[43]

to Samanta et al. [32] glucan (**1**) caused enhanced production of NO in a dose-dependent manner with optimum production of macrophages at 50 μg/mL and splenocyte and thymocyte proliferation indices were found maximum at 100 μg/mL with (**1**) [32]. Samanta et al. [32] discussed that increased NO production was the indicator of macrophage activation [32]. Therefore, this glucan (**1**) could be considered as an immunostimulator. In addition, the lipid peroxidation inhibitory activity study showed that 1 could be used as antioxidant food additives due to its EC_{50} value which is 94.32 ± 7.43 μg/mL [32].

Devi et al. [21] isolated a water soluble glucan (**2**), which exhibited enhanced immune cell activation and antitumor potential in tumor bearing mice model, from the *Pleurotus ostreatus* mushroom. Intracellular glucans isolated from *P. ostreatus* mycelia was consisted of especially fucose, mannose, and glucose with α-linkages and folded into a triple helical group. The authors also indicated that the functional property of the heteroglucan may be closely linked to its structural entities. The glucan demonstrated no cytotoxic activity against Sarcoma 180 cells. Hence, antitumor effect was immune system mediated. Since enhancing the surveillance mechanism in order to contribute to tumor inhibition the biomacromolecule glucan, an immunomodulator, from mushroom source may be used as a therapeutic agent in such cases [21].

Immunostimulating water soluble (**3**) and water insoluble (**4**) β-glucans, PS-I and PS-II, respectively, indicating important macrophage, splenocyte, and thymocyte activation, were isolated from an edible mushroom *Termitomyces robustus* var. by using hot water extraction. The isolated both glucans exhibited effective splenocyte and thymocyte stimulation at 10 μg/mL [33].

The macrophages were also activated by compound PS-I (**3**) and compound PS-II (**4**). The activation of macrophages by (**3**) and (**4**) were studied *in vitro* by NO production in the supernatant of the culture, and found that activation was in a dose-dependent manner. Stimulation of splenocytes and thymocytes is an indicator of immune activation. Several polysaccharides of type **3** as well as polysaccharide **3** also exhibited the activation, in a dose-dependent manner [33–34].

A β-glucan, insoluble in water, (**5**) was obtained from alkali extract of a somatic hybrid of *Pleurotus florida* and *Calocybe indica* var. APK2 coded PCH9FB. The glucan (**5**) also indicated splenocyte, thymocyte, and macrophage activation activities, in which glucan (**5**) can be used as a source of natural immunostimulator [35].

The alkali extract of edible *P. florida* gave two glucans (**6** and **7**) showing macrophage, splenocyte, and thymocyte activations which were performed in mouse cell culture [14]. The polysaccharide also showed *in vitro* macrophage activation [14]. The antitumor and immunomodulatory activities of polysaccharides changed with their molecular masses, their degrees of branching, conformations, and chemical modifications. The high-molecular weight polysaccharides

were more active than low-molecular weight ones. The molecular weight factor was noted to be an important criteria affecting bioactivity.

The aqueous extract of somatic hybrid (Pflo Vv5 FB) between *P. florida* and *Volvariella volvacea* strains afforded a water-soluble immunoenhancing polysaccharide (**8**). The stimulation of the macrophages, splenocytes, and thymocytes was also found by this glucan. Dose-dependent studies demonstrated that 50 μg/mL of the polysaccharide (**8**) is efficient for the splenocyte proliferator, while 80 μg/mL of the polysaccharide (**8**) is for thymocyte proliferator [36].

In the previous several studies on polysaccharides from mushrooms (*P. florida*, cultivar Assam Florida and *Termitomyces striatus*) two polysaccharides (**9** and **10**) indicating similar splenocyte and macrophage activation were isolated [37–38]. The composition of polysaccharide (**9**) was found to be D-glucose, D-galactose, D-mannose as well as L-fucose with the molar ratios of 2:1:1:1, respectively. The polysaccharide (**10**), however, was elucidated as (1→3)-, (1→6)- branched glucan.

Sarcodon aspratus (Berk.) S. Ito., an edible species, yielded a water-soluble polysaccharide (**11**) molecular mass of which was detected as 6.7 × 105 Da. Immunological activity was tested by using H3-thymidine incorporation method. Compound **11** demonstrated activation, the proliferation of the cultured mice spleen lymphocyte dose dependently. Han et al. [15] reported that polysaccharide (**11**) is one of the potential immunomodulators and could be used in foods, food additives, or medicines to protect from tumors and pathogens [15].

A glucan soluble in water (**12**) was isolated from the alkali extract of an edible mushroom *Lentinus squarrosulus* (Mont.) Singer. This glucan consisted of (1→3, 6)- linked, (1→3)-linked, (1→6)-linked, and terminal *β*-D-glucopyranosyl in a proportion of approximately 1:2:1:1. And even at 10 μM (**12**) also demonstrated activations of macrophages and of splenocytes and thymocytes [39]. Likewise, the lentinan which is also a glucan, has been shown to stimulate the immune system by activating macrophages (T-helper, NK, and other cells) and for this reason inhibits or decreases the tumor growth [40].

By using high-speed counter current chromatography polysaccharides (**13**–**15**) from *Auricularia polytricha* mushroom were obtained. Only the structure of **13** was elucidated by the researchers, while polysaccharides **14** and **15** had no definite structure supported by the data of ^{1}H and ^{13}C NMR. The structure of **13** was elucidated from partial hydrolysis; periodate oxidation, acetylation, methylation analysis, as well as NMR spectroscopy [19]. The growth inhibition of transplanted S180 sarcoma in mice was tested for these polysaccharides (**13**–**15**). While, the polysaccharide **13** demonstrated 40.4% inhibition rate, the other two polysaccharides **14** and **15** showed lower activity (12.01% and 26.15% inhibitions, respectively) in the anticancer test. In the test, cyclophosphamide had been used as a positive control and it showed a high-inhibitory rate on S180 sarcoma in mice. The mixture of those polysaccharides (**13**–**15**) also inhibited the growth of transplanted S180 sarcoma in mice with an inhibition rate of 20.03%. Among them, **13** had the highest activity. In addition, the advantage of polysaccharides isolated is that they had no toxicity [19].

A water-soluble polysaccharide (**16**) showing splenocyte activation, was obtained from the water extract of *Amorphophallus campanulatus*. The splenocyte activation tests as determined by the MTT assay were performed in mouse cell culture medium [41]. It was reported that glucan **16** contains mainly four sugars; namely, D-galactose, D-glucose, 4-*O*-acyl-D-methyl galacturonate, and L-arabinose. The molar ratios of sugars were calculated as 2-1-1-1, respectively [41].

Genus *Pleurotus*, known as oyster mushroom, provide biologically active glucans, and cultured almost all over the world. Specific glucans have been isolated from edible mushrooms *P. ostreatus* and *Pleurotus eryngii* by hot water and alkali extraction [42]. Water soluble (**17**), alkali soluble (**18**), and insoluble (**19**) fractions were obtained and characterized by different analytical techniques. The analysis showed that the fractions contained 1,3-1,6-β-D-glucan branched glucans predominated in **17** and **19**. Some proteins in complex form with glucans were also found in **17** [42].

Potential prebiotic activity of polysaccharide fractions (**17**) and (**18**) was tested using nine probiotic strains of *Lactobacillus*, *Bifidobacterium* and *Enterococcus*. The fractions (**17**) and (**18**) were subjected to synbiotic construction using some selected probiotic strains. The results of *Synytsya et al*. [42] showed that the glucan polysaccharides obtained from *P. eryngii* and *P. ostreatus* provide effects beneficial for human health [42].

In most cases, the fractions **17** and **18** from *P. ostreatus* and *P. eryngii* have been shown to contribute to growth rate of probiotic bacteria as well as the short-chain fatty acids production for *Lactobacillus* strains. The extracts from *P. eryngii* enhanced growth rate better than the extracts from *P. ostreatus* [42].

Probiotics utilization of (**17**) and (**18**) fractions from two wild edible mushrooms (*P. ostreatus* and *P. eryngii*) present various polysaccharide structures. The authors also discussed that the glucans **17** and **18** as well as proteoglucans from *Pleurotus* are useful complexes that can be used with probiotic strains for synbiotic construction. Lactobacillus strains might also be selected for synbiotic construction [42].

By using different solvent systems such as water at 120 °C, and sodium hydroxide (0.5 M) at 25 and 65 °C, and sodium phosphate buffer (0.2 M) at 25, 40, and 80 °C, six water-soluble polysaccharides (**19**–**24**) were obtained from *Ganoderma tsugae*. Polysaccharides **19** and **20** were identified as heteropolysaccharide–protein complexes, which exhibited protein contents of 13.5% and 20.1%, respectively. Polysaccharides **21** and **22** contained both α and β-D-glucans, while polysaccharides **23** and **24** were mainly (1→6)- branched (1→3)-β-D-glucan. The averages of molecular weights of the six polysaccharide samples (**19**–**24**) were found to be 62.8×10^4, 81.8×10^4, 465×10^4, 468×10^4, 176×10^4, and 161×10^4 Da, respectively. Antitumor activity was assessed by using solid tumor Sarcoma 180 cells. It was observed that the polysaccharides **19**, **20**, and **22** demonstrated antitumor activity as each indicated more than 50% inhibition [43].

The Terpenoids

Terpenes are hydrocarbons resulting from the combination of several isoprene units. Terpenoids have been associated with various pharmacological activities like anticancer activities [44–48], antimalarial [49], anticholinesterase [50], antiviral [51], antibacterial [48,52], and anti-inflammatory [53]. For example, oleanolic acid, which is the principal constituent of the triterpenoid portion of the drug, was reported for anti-inflammatory, antihyperlipidemic, and antitumor effects [54–55].

Different types of terpenes demonstrating therapeutic efficiency have been identified from mushroom species to date [56]. For example, some triterpenoids isolated from *G. lucidum* including ganoderic, lucidenic, ganodermic, ganoderenic, ganolucidic and applanoxidic acids, lucidones, ganoderals, and ganoderols have potentials for cancer disease treatments [57].

Most of the terpenoid compounds isolated from mushrooms are the lanostane-type triterpenoids. Many lanostane triterpenes have been reported as potential anticancer agents [44–48], and it has been also found that they have ability to pull in the cell cycle not only in the G2/M but also in the G1, to imprint MRP and MDP which are the transporting proteins [44]. In addition, triterpenoids can increase the levels of Bax and p53, and can decrease the matrix metallo proteinase expression [44]. Another ability of them is inhibiting the phosphorylation of Erk1/2 or inducing apoptosis as well as inhibiting the activation of NF-κB along with AP-1 [44]. In this subsection, biologically active mono-, sesqui-, di-, and tri-terpenoids which have been isolated from 29 various mushroom species have been evaluated. Briefly, from these species, 259 terpenoids have been isolated. Seven of them are monoterpenoids, 19 of them are sesquiterpenoids, 54 of them are diterpenoids, and 179 of them are triterpenoids structured. Various *in vitro* and in vivo activities of these compounds have been investigated by different research groups. The biologically active triterpenoids and their activity values were evaluated in Table 2, and the structures were given in Figure 1.

In a recent study by *Kim et al*. [44] among 15 lanostane triterpenoids (**25**−**39**), 4 of them (**25**–**28**) have been recently isolated from *Naematoloma fasciculare* species [44]. The antiproliferative activity against four human cancer cell lines; namely, colon adenocarcinoma (HCT-15), ovary malignant ascites (SK-OV-3), skin melanoma (SK-MEL-2), and nonsmall cell lung adenocarcinoma (A549) have been evaluated using the Sulforhodamine B (SRB) assay and have been performed on all of these isolated lanostane triterpenoids (**25**−**39**). Almost all compounds except compounds **25** and **26** indicated cytotoxicity against the above human cancer cells demonstrating IC_{50} (the half maximal inhibitory concentration) values between 2.29 and 28.48 μM concentrations [44].

In addition, the compounds **27** and **28** indicated important cytotoxicity. Compound **27** showed 6.59, 7.08, 8.26, and 8.53 μM IC_{50} values, against HCT-15, A549, SK-MEL-2, SK-OV-3, respectively. Compound **28**, however, demonstrated 3.99, 7.36, 4.77, and 8.50 μM IC_{50} values, against same cancer cell lines,

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
Naematoloma fasciculare	**25**	*Cytotoxic activity*: IC_{50} > 30 μM against HCT-15, A549, SK-MEL-2, and SK-OV-3	[44]
N. fasciculare	**26**	*Cytotoxic activity*: IC_{50} > 30 μM against HCT-15, A549, SK-MEL-2, and SK-OV-3	[44]
N. fasciculare	**27**	*Cytotoxic activity*: IC_{50}: 6.59 μM against HCT-15 IC_{50}: 7.08 μM against A549 IC_{50}: 8.26 μM against SK-MEL-2 IC_{50}: 8.53 μM against SK-OV-3	[44]
N. fasciculare	**28**	*Cytotoxic activity*: IC_{50}: 3.99 μM against HCT-15 IC_{50}: 7.36 μM against A549 IC_{50}: 4.77 μM against SK-MEL-2 IC_{50}: 8.50 μM against SK-OV-3	[44]
N. fasciculare	**29**	*Cytotoxic activity*: IC_{50}: 8.60 μM against SK-MEL-2 IC_{50} > 30 μM against HCT-15, A549 and SK-OV-3	[44]
N. fasciculare	**30**	*Cytotoxic activity*: IC_{50}: 9.06 μM against SK-MEL-2 IC_{50} > 30 μM against HCT-15, A549 and SK-OV-3	[44]
N. fasciculare	**31**	*Cytotoxic activity*: IC_{50}: 28.48 μM against HCT-15 IC_{50}: 27.20 μM against A549 IC_{50}: 16.60 μM against SK-MEL-2 IC_{50}: 22.17 μM against SK-OV-3	[44]
N. fasciculare	**32**	*Cytotoxic activity*: IC_{50}: 27.96 μM against HCT-15 IC_{50}: 25.52 μM against A549 IC_{50}: 25.23 μM against SK-MEL-2 IC_{50}: 16.17 μM against SK-OV-3	[44]
N. fasciculare	**33**	*Cytotoxic activity*: IC_{50}: 9.16 μM against SK-MEL-2 IC_{50} > 30 μM against HCT-15, A549 and SK-OV-3	[44]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
N. fasciculare	**34**	*Cytotoxic activity*: IC_{50}: 5.73 μM against SK-MEL-2 IC_{50} > 30 μM against HCT-15, A549 and SK-OV-3	[44]
N. fasciculare	**35**	*Cytotoxic activity*: IC_{50}: 7.85 μM against HCT-15 IC_{50}: 8.53 μM against A549 IC_{50}: 5.17 μM against SK-MEL-2IC_{50}: 8.22 μM against SK-OV-3	[44]
N. fasciculare	**36**	*Cytotoxic activity*: IC_{50}: 2.37 against HCT-15 IC_{50}: 2.82 μM against A549 IC_{50}: 2.29 μM against SK-MEL-2 IC_{50}: 3.06 μM against SK-OV-3	[44]
N. fasciculare	**37**	*Cytotoxic activity*: IC_{50}: 9.86 μM against the SK-OV-3 IC_{50} > 30 μM against HCT-15, A549 and SK-MEL-2	[44]
N. fasciculare	**38**	*Cytotoxic activity*: IC_{50}: 7.98 μM against the SK-OV-3 IC_{50} > 30 μM against HCT-15, A549 and SK-MEL-2	[44]
N. fasciculare	**39**	*Cytotoxic activity*: IC_{50}: 4.47 against HCT-15 IC_{50}: 3.29 μM against A549 IC_{50}: 4.54 μM against SK-MEL-2 IC_{50}: 7.71 μM against SK-OV-3	[44]
Pleurotus cornucopiae (Paulet) Rolland (Pleurotaceae)	**40**	*NO production inhibitory activity*: IC_{50}: 6.8 μM in RAW 264.7 *Cytotoxic activity*: IC_{50}: 20.6 against HeLa	[45]
P. cornucopiae (Paulet) Rolland (Pleurotaceae)	**41**	*NO production inhibitory activity*: IC_{50}: 12.6 μM in RAW 264.7 *Cytotoxic activity*: IC_{50}: 32.8 against HeLa	[45]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
P. cornucopiae (Paulet) Rolland (Pleurotaceae)	**42**	*NO production inhibitory activity*: IC_{50}: 20.8 μM in RAW 264.7 *Cytotoxic activity*: IC_{50}: 18.8 against HeLa	[45]
P. cornucopiae	**46**	*NO production inhibitory activity*: IC_{50}: 81.8 μM on macrophage RAW 264.7	[46]
P. cornucopiae	**47**	*NO production inhibitory activity*: IC_{50}: 88.8 μM on macrophage RAW 264.7	[46]
P. cornucopiae	**48**	*NO production inhibitory activity*: IC_{50}: 80.4 μM on macrophage RAW 264.7	[46]
P. cornucopiae	**49**	*NO production inhibitory activity*: IC_{50}: 65.6 μM on macrophage RAW 264.7	[46]
P. cornucopiae	**50**	*NO production inhibitory activity*: IC_{50}: 72.8 μM on macrophage RAW 264.7	[46]
P. cornucopiae	**51**	*Growth inhibitory activity*: IC_{50}: 76.8 μM against HepG2 *Cytotoxic activity*: IC_{50}: 70.6 μM against HeLa	[46]
P. cornucopiae	**52**	*NO production inhibitory activity*: IC_{50}: 76.5 μM on macrophage RAW 264.7 *Growth inhibitory activity*: IC_{50}: 68.6 μM against HepG2 *Cytotoxic activity*: IC_{50}: 36.0 μM against HeLa	[46]
P. cornucopiae	**53**	*NO production inhibitory activity*: IC_{50}: 72.4 μM on macrophage RAW 264.7	[46]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
Pleurotus eryngii	**54**	*Cytotoxic activity*: IC_{50}: 20.6 μM against HeLa IC_{50}: 28.6 μM against HepG2	[47]
P. eryngii	**55**	*Cytotoxic activity*: No activity at 100 μM against HeLa and HepG2	[47]
P. eryngii	**56**	*Cytotoxic activity*: No activity at 100 μM against HeLa and HepG2	[47]
Astraeus odoratus	**57**	*Cytotoxic activity*: No activity against KB, NCI-H187 and MCF-7 *Antimycobacterial activity*: No activity against *M. tuberculosis* $H_{37}Ra$	[48]
A. odoratus	**58**	*Cytotoxic activity*: IC_{50}: 34.69 μg/mL against KB IC_{50}: 19.99 μg/mL against NCI-H187 No activity against MCF-7 *Antimycobacterial activity*: MIC: 50 μg/mL against *M. tuberculosis* $H_{37}Ra$	[48]
A. odoratus	**59**	*Cytotoxic activity*: IC_{50}: 18.57 μg/mL against KB IC_{50}: 48.35 μg/mL against NCI-H187 No activity against MCF-7 *Antimycobacterial activity*: MIC: 25 μg/mL against *M. tuberculosis* $H_{37}Ra$	[48]
A. odoratus	**60**	*Cytotoxic activity*: No activity against KB, NCI-H187 and MCF-7 *Antimycobacterial activity*: No activity against *M. tuberculosis* $H_{37}Ra$	[48]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
A. odoratus	**61**	*Cytotoxic activity*: IC_{50}: 31.55 μg/mL against KB, IC_{50}: 34.15 μg/mL against NCI-H187 IC_{50}: 40.15 μg/mL against MCF-7 *Antimycobacterial activity*: No activity against *M. tuberculosis* $H_{37}Ra$	[48]
Astraeus hygrometricus	**62**	*Anticandidal activity*: MIC: 0.25 μg/disc against *C. albicans*	[58]
A. hygrometricus	**63**	*Anticandidal activity*: MIC: 0.20 μg/disc against *C. albicans*	[58]
Astraeus pteridis	**64**	*Antimycobacterial activity*: MIC: 58.0 μg/mL against *M. tuberculosis* $H_{37}Rr$ *Cytotoxic activity*: No activity against Vero cells (African green monkey kidney fibroblasts)	[59]
A. pteridis	**65**	*Antimycobacterial activity*: MIC: 64.0 μg/mL against *M. tuberculosis* $H_{37}Rr$ *Cytotoxic activity*: No activity against Vero cells	[59]
A. pteridis	**66**	*Antimycobacterial activity*: MIC > 64.0 μg/mL against *M. tuberculosis* $H_{37}Rr$ *Cytotoxic activity*: No activity against Vero cells	[59]
A. pteridis	**67**	*Antimycobacterial activity*: MIC > 64.0 μg/mL against *M. tuberculosis* $H_{37}Rr$ *Cytotoxic activity*: No activity against Vero cells	[59]
A. pteridis	**68**	*Antimycobacterial activity*: MIC: 34.0 μg/mL against *M. tuberculosis* $H_{37}Rr$ *Cytotoxic activity*: No activity against Vero cells	[59]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
Ganoderma orbiforme	**69**	*Antimycobacterial activity*: MIC > 89 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50} > 18 μM against *Plasmodium falciparum* K1 *Cytotoxic activity*: IC_{50} > 89 μM against NCI-H187, MCF-7, KB and Vero cells	[49]
G. orbiforme	**72**	*Antimycobacterial activity*: MIC > 85 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50} > 17 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50} > 85 μM against NCI-H187, MCF-7, KB and Vero cells	[49]
G. orbiforme	**73**	*Antimycobacterial activity*: MIC > 95 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50} > 19 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50}: 70 μM against NCI-H187 IC_{50} > 95 μM against MCF-7, KB and Vero cells	[49]
G. orbiforme	**74**	*Antimycobacterial activity*: MIC: 96 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50} > 19 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50}: 44 μM against NCI-H187 IC_{50} > 96 μM against MCF-7 IC_{50}: 63 μM against KB IC_{50}: 36 μM against Vero cells	[49]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. orbiforme	**75**	*Antimycobacterial activity*: MIC > 98 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50} > 20 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50}: 65 μM against NCI-H187 IC_{50} > 98 μM against MCF-7 IC_{50}: 65 μM against KB IC_{50}: 35 μM against Vero cells	[49]
G. orbiforme	**80**	*Antimycobacterial activity*: MIC > 95 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50} > 19 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50}: 34 μM against NCI-H187 IC_{50} > 95 μM against MCF-7, KB and Vero cells	[49]
G. orbiforme	**81**	*Antimycobacterial activity*: MIC: 1.3 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50}: 4.6 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50}: 13 μM against NCI-H187 IC_{50}: 51 μM against MCF-7 IC_{50}: 13 μM against KB IC_{50}: 16 μM against Vero cells	[49]
G. orbiforme	**83**	*Antimycobacterial activity*: MIC: 10 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50}: 5.5 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50}: 15 μM against NCI-H187 IC_{50}: 78 μM against MCF-7 IC_{50}: 18 μM against KB IC_{50}: 28 μM against Vero cells	[49]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. orbiforme	**85**	*Antimycobacterial activity*: MIC > 88 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50}: 17 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50}: 26 μM against NCI-H187 IC_{50}: 58 μM against MCF-7 IC_{50}: 40 μM against KB IC_{50} > 88 μM against Vero cells	[49]
G. orbiforme	**87**	*Antimycobacterial activity*: MIC: 98 μM against *M. tuberculosis* $H_{37}Ra$ *Antimalarial activity*: IC_{50} > 20 μM against *P. falciparum* K1 *Cytotoxic activity*: IC_{50}: 39 μM against NCI-H187 IC_{50} > 98 μM against MCF-7 IC_{50}: 53 μM against KB IC_{50} > 98 μM against Vero cells	[49]
Ganoderma lucidum	**89**	*Anticholinesterase activity*: IC_{50}: 18.35 ± 2.57 μM against AChE	[50]
G. lucidum	**90**	*Anticholinesterase activity*: IC_{50}: 9.40 ± 0.88 μM against AChE	[50]
G. lucidum	**91**	*Anticholinesterase activity*: IC_{50}: 30.26 ± 1.39 μM against AChE	[50]
		Effects on adipocyte differentiation: 62% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**92**	*Anticholinesterase activity*: IC_{50}: 22.63 ± 1.36 μM against AChE	[50]
		Effects on adipocyte differentiation: 78% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. lucidum	**93**	*Anticholinesterase activity*: IC_{50}: 18.35 ± 2.95 μM against AChE	[50]
		Effects on adipocyte differentiation: 80% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**94**	*Anticholinesterase activity*: IC_{50}: 22.97 ± 2.24 μM against AChE	[50]
		Effects on adipocyte differentiation: 79% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**95**	*Anticholinesterase activity*: IC_{50}: 26.78 ± 0.50 μM against AChE	[50]
G. lucidum	**96**	*Anticholinesterase activity*: IC_{50}: 21.11 ± 1.66 μM against AChE	[50]
G. lucidum	**97**	*Anticholinesterase activity*: IC_{50}: 21.84 ± 2.29 μM against AChE	[50]
G. lucidum	**98**	*Anticholinesterase activity*: IC_{50}: 16.27 ± 0.51 μM against AChE	[50]
Ganoderma pfeifferi	**98**	*Antiviral activity*: ED_{50}: 0.068 μM against Herpes simplex ED_{50}: 0.22 μM against Influenza A	[51]
G. lucidum	**99**	*Anticholinesterase activity*: IC_{50}: 25.05 ± 2.48 μM against AChE	[50]
G. lucidum	**100**	*Anticholinesterase activity*: IC_{50}: 22.31 ± 0.92 μM against AChE	[50]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. lucidum	**101**	*Anticholinesterase activity*: IC_{50}: 31.03 ± 1.69 μM against AChE IC_{50}: 156.27 ± 6.12 μM against BChE	[50]
G. lucidum	**102**	*Anticholinesterase activity*: IC_{50}: 25.91 ± 0.89 μM against AChE IC_{50}: 188.36 ± 3.05 μM against BChE *Effects on adipocyte differentiation*: 79% lipid accumulation at 40 μg/mL in 3T3-L1 cells *Anti-invasive Activity*: 98 ± 4% at 50 μM against $HepG_2$	[50]
G. lucidum	**103**	*Anticholinesterase activity*: IC_{50}: 24.04 ± 3.46 μM against AChE	[50]
		Anti-invasive Activity: 99 ± 5% at 50 μM against $HepG_2$	[63]
G. lucidum	**104**	*Anticholinesterase activity*: IC_{50}: 17.14 ± 2.88 μM against AChE	[50]
		Effects on adipocyte differentiation: 76% Lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. pfeifferi		*Antiviral activity*: No activity against Herpes simplex ED_{50}: 0.22 μM against Influenza A	[51]
G. lucidum	**105**	*Anticholinesterase activity*: IC_{50}: 11.58 ± 0.36 μM against AChE	[50]
		Effects on adipocyte differentiation: 43% Lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. lucidum	**106**	*Anticholinesterase activity*: IC_{50}: 12.26 ± 0.68 μM against AChE	[50]
		Effects on adipocyte differentiation: 57% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**107**	*Effects on adipocyte differentiation*: 52% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**108**	*Effects on adipocyte differentiation*: 59% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**109**	*Effects on adipocyte differentiation*: 79% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**110**	*Effects on adipocyte differentiation*: 76% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**111**	*Effects on adipocyte differentiation*: 77% Lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**112**	*Effects on adipocyte differentiation*: 80.5% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**113**	*Effects on adipocyte differentiation*: 78.5% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**114**	*Effects on adipocyte differentiation*: 78% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. lucidum	**115**	*Effects on adipocyte differentiation*: 83% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**116**	*Effects on adipocyte differentiation*: 78% lipid accumulation at 40 μg/mL in 3T3-L1 cells	[61]
G. lucidum	**117**	*Anticomplement activity*: IC_{50}: 44.6 μM	[62]
G. lucidum	**118**	*Anticomplement activity*: $IC_{50} > 200$ μM	[62]
G. lucidum	**119**	*Anti-invasive Activity*: 100 ± 6% at 50 μM against $HepG_2$	[63]
G. lucidum	**120**	*Anti-invasive Activity*: 96 ± 5% at 50 μM against $HepG_2$	[63]
G. lucidum	**121**	*Inhibitory activity*: IC_{50}: 22.3 μM against HMG-CoA reductase IC_{50}: 5.5 μM against acyl CoA acyltransferase	[64]
G. lucidum	**122**	*Inhibitory activity*: IC_{50}: 21.7 μM against HMG-CoA reductase IC_{50}: 47.3 μM against acyl CoA acyltransferase	[64]
G. lucidum	**124**	*Cytotoxic activity*: ED_{50}: 10.4 μg/mL against Meth-A $ED_{50} > 20$ μg/mL against T-47D, LLC, and sarcoma 180	[65]
G. lucidum	**125**	*Cytotoxic activity*: ED_{50}: 15.0 μg/mL against T-47D ED_{50}: 14.3 μg/mL against LLC ED_{50}: 4.0 μg/mL against Meth-A $ED_{50} > 20$ μg/mL against sarcoma 180	[65]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. lucidum	**126**	*Cytotoxic activity*: ED_{50}: 10.7 μg/mL against T-47D ED_{50}: 4.7 μg/mL against LLC ED_{50}: 3.8 μg/mL against Meth-A ED_{50}: 7.1 μg/mL against sarcoma 180	[65]
G. lucidum	**127**	*Cytotoxic activity*: ED_{50}: 4.8 μg/mL against T-47D ED_{50} > 20 μg/mL against LLC ED_{50}: 2.8 μg/mL against Meth-A ED_{50}: 10 μg/mL against sarcoma 180	[65]
Ganoderma amboinenese	**130**	*Toxic activity*: 70.3% against brine shrimp larvae	[67]
G. amboinenese	**131**	*Toxic activity*: 75.8% against brine shrimp larvae	[67]
G. amboinenese	**142**	*Cytotoxic activity*: IC_{50}: 20.3 μg/mL against HuH-7 IC_{50}: 38.3 μg/mL against HCT-116 IC_{50}: 39.2 μg/mL against Raji IC_{50}: 26.5 μg/mL against HL60	[57]
Ganoderma colossum	**143**	*Anti-HIV-1 protease activity*: IC_{50}: 5.0 μg/mL against HIV-1 protease	[69]
G. colossum	**144**	*Anti-HIV-1 protease activity*: IC_{50} > 100.0 μg/mL against HIV-1 protease	[69]
G. colossum	**145**	*Anti-HIV-1 protease activity*: IC_{50}: 8.0 μg/mL against HIV-1 protease	[69]
G. colossum	**146**	*Anti-HIV-1 protease activity*: IC_{50}: 31.4 μg/mL against HIV-1 protease	[69]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. colossum	**147**	*Anti-HIV-1 protease activity*: IC_{50}: 9.0 μg/mL against HIV-1 protease	[69]
G. colossum	**148**	*Anti-HIV-1 protease activity*: IC_{50}: 13.8 μg/mL against HIV-1 protease	[69]
G. colossum	**149**	*Anti-HIV-1 protease activity*: IC_{50}: 5.0 μg/mL against HIV-1 protease	[69]
G. colossum	**150**	*Anti-HIV-1 protease activity*: IC_{50}: 39.0 μg/mL against HIV-1 protease	[69]
G. pfeifferi	**152**	*Antiviral activity*: IC_{50}: 0.3 μg/mL against Herpes simplex No activity against Influenza A	[70]
G. pfeifferi	**153**	*Antiviral activity*: No activity against Herpes simplex IC_{50}: 2.6 μg/mL against Influenza A	[70]
G. pfeifferi	**154**	*Antiviral activity*: IC_{50}: 0.075 μg/mL against Herpes simplex IC_{50}: 3.0 μg/mL against Influenza A	[70]
G. pfeifferi	**155**	*Antiviral activity*: IC_{50}: 0.75 μg/mL against Herpes simplex No activity against Influenza A	[70]
G. pfeifferi	**156**	*Antiviral activity*: IC_{50}: 0.03 μg/mL against Herpes simplex No activity against Influenza A	[70]
G. pfeifferi	**157**	*Antiviral activity*: No activity against Herpes simplex ED_{50}: 0.19 μM against Influenza A	[51]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
Ganoderma fornicatum	**158**	*Inhibitory effects on rabbit platelet aggregation*: 41.2 ± 1% at 4.5 nmol on PAF-induced 45.0 ± 7% at 5.0 μmol on ADP-induced 65.2 ± 7.6% at 240.0 μmol on AA-induced	[71]
G. fornicatum	**159**	*Inhibitory effects on rabbit platelet aggregation*: 44.3 ± 1.7% at 4.5 nmol on PAF-induced 58.3 ± 7.9% at 5.0 μmol on ADP-induced 65.4 ± 5.4% at 240.0 μmol on AA-induced	[71]
Ganoderma concinna	**161**	*Cytotoxic activity*: IC_{50} > 100 μM against HL-60	[72]
G. concinna	**162**	*Cytotoxic activity*: IC_{50}: 15.7 ± 4.6 μM against HL-60	[72]
G. concinna	**163**	*Cytotoxic activity*: IC_{50} > 100 μM against HL-60	[72]
Ganoderma tsugae	**170**	*Cytotoxic activity*: ED_{50}: 6.5 μg/mL against PLC/PRF/5 ED_{50}: 8.6 μg/mL against T-24 ED_{50}: no activity against 212 ED_{50}: 7.2 μg/mL against HT-3 ED_{50}: 9.5 μg/mL against SiHa	[75]
G. tsugae	**171**	*Cytotoxic activity*: ED_{50}: 6.8 μg/mL against PLC/PRF/5 ED_{50}: 3.1 μg/mL against T-24 ED_{50}: no activity against CaSKi	[75]
G. tsugae	**172**	*Cytotoxic activity*: ED_{50}: no activity against PLC/PRF/5 ED_{50}: no activity against T-24 ED_{50}: 10.3 against 212 ED_{50}: no activity against CaSKi	[75]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
G. tsugae	**173**	*Cytotoxic activity*: ED_{50}: 1.73 μg/mL against PLC/PRF/5 ED_{50}: no activity against 212 ED_{50}: 6.8 μg/mL against HT-3 ED_{50}: 8.4 μg/mL against SiHa	[75]
G. tsugae	**174**	*Cytotoxic activity*: ED_{50}: 4.4 μg/mL against T-24 ED_{50}: 3.5 μg/mL against HT-3 ED_{50}: 5.5 μg/mL against SiHa ED_{50}: 6.2 μg/mL against CaSKi	[75]
G. tsugae	**175**	*Cytotoxic activity*: ED_{50}: 12.1 μg/mL against PLC/PRF/5 ED_{50}: 8.2 μg/mL against T-24 ED_{50}: 6.9 μg/mL against HT-3 ED_{50}: 5.1 μg/mL against SiHa ED_{50}: 7.2 μg/mL against CaSKi	[75]
Russula lepida and *Russula amarissima*	**189**	*Cell growth inhibitory activity* IC_{50}: 72 μM against WISH IC_{50}: 85.4 μM against CAKI 1 IC_{50}: 90.3 μM against A549	[76]
Sarcodon scabrosus	**195**	*Cytotoxic activity*: IC_{50}: 20 μM against HeLa	[80]
		Antibacterial activity: 6.5 mm zone against *Bacillus subtilis* 7.5 mm zone against *Staphylococcus aureus*	[52]
S. scabrosus	**203**	*Anti-inflammatory activity*: 75% at 630 mmol in the mause ear	[53]
S. scabrosus	**204**	*Anti-inflammatory activity*: 36% at 630 mmol in the mause ear *NO production inhibitory activity*: IC_{50}: 12.06 μM in RAW 264.7	[53]
Cyathus africanus		*Cytotoxic activity*: IC_{50}: 5.86 against HeLa IC_{50}: 3.68 against K562	[94]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
S. scabrosus	**205**	*Anti-inflammatory activity*: 61% at 630 mmol in the mause ear	[53]
S. scabrosus	**206**	*Anti-inflammatory activity*: 46% at 630 mmol in the mause ear	[53]
S. scabrosus	**207**	*Anti-inflammatory activity*: 72% at 630 mmol in the mause ear	[53]
S. scabrosus	**208**	*Anti-inflammatory activity*: 29% at 630 mmol in the mause ear	[53]
S. scabrosus	**209**	*Anti-inflammatory activity*: 49% at 630 mmol in the mause ear	[53]
S. scabrosus	**210**	*Anti-inflammatory activity*: 78% at 630 mmol in the mause ear	[53]
S. scabrosus	**211**	*Anti-inflammatory activity*: 53% at 630 mmol in the mause ear	[53]
S. scabrosus	**212**	*Anti-inflammatory activity*: 58% at 630 mmol in the mause ear	[53]
S. scabrosus	**213**	*Anti-inflammatory activity*: 53% at 630 mmol in the mause ear	[53]
S. scabrosus	**214**	*Anti-inflammatory activity*: 43% at 630 mmol in the mause ear	[53]
Inonotus obliquus	**233**	*Cytotoxic acitivity*: IC_{50}: 62.5 μM against L1210	[86]
I. obliquus	**235**	*Cytotoxic acitivity*: IC_{50}: 34.4 μM against L1210	[86]
I. obliquus	**237**	*Cytotoxic acitivity*: IC_{50}: 81.2 μM against A549 IC_{50}: 87.5 μM against HL60	[86]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
Spongiporus leucomallellus	**240**	*Antimicrobial activity*: 10 mm zone against *Bacillus subtilis* 11 mm zone against *Bacillus brevis* 12 mm zone against *Paecilomyces variotii* 13 mm zone against *Proteus vulgaris*	[87]
Leucopaxillus gentianeus	**242**	*Antiproliferative activity*: IC_{50}: 0.011 μg/mL against the NCI-H460	[88]
L. gentianeus	**243**	*Antiproliferative activity*: $IC_{50} > 30$ μg/mL against the NCI-H460	[88]
L. gentianeus	**244**	*Antiproliferative activity*: IC_{50}: 3.5 μg/mL against the NCI-H460 *Cell growth inhibitory activity*: $IC_{50} > 300$ μM against the A549 IC_{50}: 107.8 μM against the CAKI 1 IC_{50}: 96.7 μM against the HepG2 IC_{50}: 7.94 μM against the MCF-7	[88] [89]
L. gentianeus	**245**	*Antiproliferative activity*: IC_{50}: 0.3 μg/mL against the NCI-H460	[88]
		Cell growth inhibitory activity: IC_{50}: 154.7 μM against the A549 IC_{50}: 30.3 μM against the CAKI 1 IC_{50}: 86.9 μM against the HepG2 IC_{50}: 86.9 μM against the MCF-7	[89]
L. gentianeus	**246**	*Antiproliferative activity*: IC_{50}: 0.12 μg/mL against the NCI-H460	[88]
L. gentianeus	**247**	*Antiproliferative activity*: IC_{50}: 0.06 μg/mL against the NCI-H460	[88]

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
L. gentianeus	**248**	*Antiproliferative activity*: IC_{50}: 10.6 μg/mL against the NCI-H460	[88]
L. gentianeus	**249**	*Cell growth inhibitory activity*: IC_{50}: 46.6 μM against the A549 IC_{50}: 7.1 μM against the CAKI 1 IC_{50}: 0.76 μM against the HepG2 IC_{50}: 0.78 μM against the MCF-7	[89]
L. gentianeus	**250a-c**	*Cell growth inhibitory activity*: $IC_{50} > 300$ μM against the A549, CAKI 1, HepG2, MCF-7	[89]
Hebeloma versipelle	**251**	*Cytotoxic activity*: IC_{50}: 11.2 μg/mL against HL60 IC_{50}: 20.9 μg/mL against Bel-7402 IC_{50}: 22.6 μg/mL against SGC-7901 IC_{50}: 25.0 μg/mL against A549	[90]
Tricholoma saponaceum	**252**	*Cytotoxic activity*: IC_{50}: 8.9 μM against HL-60	[91]
Elfvingia applanata	**263**	*Cytotoxic activity*: IC_{50}: 1.1 μg/mL against Ehlrich and Kato III cells	[92]
Hypholoma fasciculare	**264**, **265**, **266**	*Antimicrobial activity*: IC_{50}: 14.0 μM against 3α-hydroxysteroid dehydrogenase	[93]
C. africanus	**267**	*NO production inhibitory activity*: IC_{50}: 71.99 μM in RAW 264.7 *Cytotoxic activity*: No activity against HeLa and K562	[94]
C. africanus	**268**	*NO production inhibitory activity*: IC_{50}: 89.43 μM in RAW 264.7 *Cytotoxic activity*: No activity against HeLa $IC_{50} > 100$ against K562	[94]

Continued

TABLE 2 The Biologically Active Triterpenoids and Their Activity Values—Continued

Mushroom Species	Isolated Terpenoids	Bioactivities Tested with Activity Values	References
C. africanus	**269**	*NO production inhibitory activity*: IC_{50}: 1.45 μM in RAW 264.7 *Cytotoxic activity*: IC_{50} > 100 against HeLa IC_{50} > 100 against K562	[94]
C. africanus	**270**	*NO production inhibitory activity*: IC_{50}: 84.16 μM in RAW 264.7 *Cytotoxic activity*: IC_{50}: 23.72 against HeLa IC_{50}:39.46 against K562	[94]
C. africanus	**271**	*NO production inhibitory activity*: IC_{50}: 12.0 μM in RAW 264.7 *Cytotoxic activity*: No activity against HeLa and K562	[94]
C. africanus	**272**	*NO production inhibitory activity*: IC_{50}: 55.70 μM in RAW 264.7 *Cytotoxic activity*: IC_{50} > 100 against HeLa IC_{50} > 100 against K562	[94]
C. africanus	**273**	*NO production inhibitory activity*: IC_{50}: 10.73 μM in RAW 264.7 *Cytotoxic activity*: IC_{50}: 5.68 against HeLa IC_{50}: 8.64 against K562	[94]
C. africanus	**274**	*NO production inhibitory activity*: IC_{50}: 9.45 μM in RAW 264.7 *Cytotoxic activity*: No activity against HeLa IC_{50} > 100 against K562	[94]
C. africanus	**275**	*NO production inhibitory activity*: IC_{50}: 47.36 μM in RAW 264.7 *Cytotoxic activity*: IC_{50} > 100 against HeLa IC_{50} > 100 against K562	[94]

respectively. Moreover, compounds **35**, **36**, and **39** exhibited moderate cytotoxicity against same cancer cell lines. IC_{50} values of compound (**35**) found to be 7.85, 8.53, 5.17, and 8.22 μM, IC_{50} values of compound (**36**) were 2.37, 2.82, 2.29, and 3.06 μM, and IC_{50} values of compound (**39**) were 4.47, 3.29, 4.54, and 7.71 μM, respectively [44].

Interestingly, triterpenes **29**, **30**, **33**, and **34**, have a side chain of N-glycyl-3-hydroxy-3-methylglutaryl group in C-2 and C-3 positions. These compounds exhibited significant selective cytotoxic activity against human melanoma (SK-MEL-2) cell line with IC_{50} values of 8.60, 9.06, 9.16, and 5.73 μM [44].

The position of side chain in the triterpenoid sometimes affects the cytotoxic activity. The compounds **25** and **31** or **26** and **32** were the examples of this theory. The difference between compounds **25** and **31** or compounds **26** and **32** is the position of the 3-hydroxy-3-methylglutaryl group. As mentioned in the paper by Kim et al. [44], if 3-hydroxy-3-methylglutaryl group bonded at C-3 position such as in compounds **25** and **26** it reduced the activity substantially. In contrast, like in compounds **31** and **32**, when bonded at C-2 position the cytotoxic activity increased [44].

Compounds **37** and **38** having ketone functionalities at C-12 have demonstrated selective cytotoxicity with IC_{50} values of 9.86 and 7.98 μM against the SK-OV-3 cell line [44].

Pleurotus cornucopiae (Paulet) Rolland (Pleurotaceae) is an edible mushroom and consumed in China, frequently. Edible mushroom may be a source of bioactive metabolites to develop new drugs and new nutraceuticals. The chromatographic studies of this mushroom gave five benzannulated 5,5-spiroketals, named, pleurospiroketals A−E (**40**−**44**), and the compound (6S,7S)-6,7-dihydroxy-3,6-dimethyl-2-isovaleroyl-4,5,6,7-tetrahydrobenzofuran (**45**) [45].

Pleurospiroketals A−E (**40**−**44**) are rare natural sesquiterpenoids having a unique benzannulated 5,5-spiroketal skeleton. When the pleurospiroketals A−C (**40**−**42**) were investigated for their nitric oxide production on the macrophage cell RAW 264.7 which was activated with lipopolysaccharide, no cytotoxic activity was observed at 100 μM concentration. However, investigation of their cytotoxicity against HeLa cell lines indicated that the compounds **40**−**42** exhibited NO inhibitory activity demonstrating IC_{50} values of 6.8, 12.6, and 20.8 μM, respectively. According to Wang et al. [45], the exocyclic double bond at C-3 position in **40** and **41** enhances the nitric oxide (NO) inhibitory activity. As known, overproduction of NO is related with various disorders, including inflammatory autoimmune diseases and cancer. In addition, compounds **40**−**42** exhibited cytotoxic activity against the HeLa cancer cell line by demonstrating IC_{50} of 20.6, 32.8, and 18.8 μM, respectively [45].

The same group also isolated five monoterpenoids (**46**−**50**) and three sesquiterpenoid (**51**–**53**) where **51** being the new sesquiterpenoid, from the mycelia's of *P. cornucopiae* that was fermented on rice [46]. Compounds **46**−**53** were screened for their NO production inhibitory activity of macrophage RAW 264.7 and the growth inhibition against HeLa and HepG2 cell lines by the MTT assay.

At the concentration of 100 μM, the compounds **46**−**53** showed no cytotoxic activity to macrophage cells RAW 264.7. Compounds **46**−**50**, **52**, and **53** demonstrated inhibitory activity with IC_{50} values in the range of 60–90 μM against nitric oxide production in lipopolysaccaride-activated macrophages [46]. Compounds **51** and **52** indicated weak growth inhibitory activity against HepG2 (IC_{50}: 76.8 and 68.6 μM) and against HeLa cell lines (IC_{50}: 70.6 and 36.0 μM) in the growth inhibition assay. According to Wang et al. [46], the compounds **51** and **52** behave dissimilar impacts on HeLa, RAW 264.7, and HepG2 cells [46].

Eryngiolide A (**54**), the first member of C20 diterpenoids and having two γ-lactone units, was obtained from *P. eryngii* [47]. Cytotoxic effects of compounds **54**–**56** were tested against HeLa and HepG2 which are human cancer cell lines by using MTT assay. Only, **54** showed toxicities against both lines, indicating 20.6 and 28.6 μM IC_{50} values, respectively. In contrast, **55** and **56** exhibited no inhibitory activity [47].

Arpha et al. [48] isolated five lanostane-type triterpenes, artabotryols A (**57**), and astraodoric acids A−D (**58**–**61**) from the *Astraeus odoratus*. The isolated compounds (**57**–**61**) were investigated for their antimicrobial activity against Mycobacterium tuberculosis H37Ra, and the cytotoxicity against KB, NCI-H187, and MCF-7 cancerous cell lines [48].

Antibacterial and cytotoxic activity results showed that astraodoric acids A (**58**) and B (**59**) exhibited moderate antibacterial activity indicating MIC (Minimum Inhibitory Concentration) values of 50 and 25 μg/mL, respectively. The same compounds also showed cytotoxic activities with IC_{50} values of 34.69 and 18.57 μg/mL against KB cancer cell line, respectively. Against NCI-H187 cell line, however, **58** and **59** exhibited 19.99 and 48.35 μg/mL IC_{50} values, respectively. Although, astraodoric acid D (**61**) had no activity against *M. tuberculosis* H37Ra; but, it was found to be slightly cytotoxic against KB, NCI-H187, and MCF-7 cancer cell lines with IC_{50} values of 31.55, 34.15, and 40.15 μg/mL, respectively [48].

Two new lanostane-type triterpenes, astrakurkurol (**62**) and astrakurkurone (**63**), were isolated from *Astraeus hygrometricus*, another genus of *Astraeus*. Antifungal activity against *Candida albicans* and leishmanicidal activity against *Leishmania donovani* of these compounds were also performed by the authors [58]. The agar disc diffusion method was chosen to perform the anticandidal activity of the compounds against *C. albicans* MTCC 183. As known *C. albicans* is a virulent strain, and resistant to nystatin (20 μg/disc), fluconazole (10 μg/disc), and amphotericin-B (20 μg/disc) which are the antifungal agents used in pharmaceuticals [58].

The triterpenes demonstrated strong activity against *C. albicans* which presented MIC values between 0.25 and 0.20 μg/disc. The results were also compared against those of antibiotics noted above. The results indicated that both compounds might show potential as new antifungal drugs. In addition, the astrakurkurone (**63**) also inhibited the growth of *Leishmania donovani promastigotes in vitro*. Thus, the structures of triterpenes are advised to be considered as template structures for new compounds investigations for both anticandidal and leishmanicidal activities [58].

Another species of *Astraeus* genus was studied by Stanikunaite et al. [59]. These workers reported that the bioactivity guided fractionation of the ethyl acetate extract of *Astraeus pteridis*, the truffle-mimiking mushroom, gave five lanostane type triterpenoids; namely, 3-*epi*-astrahygrol (**64**), astrahygrone (**65**), astrapteridone (**66**), astrapteridiol (**67**), and 3-*epi*-astrapteridiol (**68**). The compounds were tested for their antimycobacterial activity against *M. tuberculosis*. Although compounds **64** and **65** showed mild activity, demonstrating MIC values of 34.0, and 58.0 μg/mL, respectively, compounds **66** and **67** showed almost no activity indicating more than 64 μg/mL MIC values. Additionally these compounds exhibited no cytotoxic activity to nonmalignant/noncancerous Vero cells (African green monkey kidney fibroblasts) at concentrations up to 100 μg/mL [59].

In a recent study, seven novel lanostane-type triterpenoids called ganorbiformins A, B, C, D, E, F, and G (**69**–**75**) and 12 known triterpenoids (**76**–**87**) were isolated from *Ganoderma orbiforme* BCC 22,324 which is a cultured mushroom [59]. The C-3 epimer of ganoderic acid T (**81**) exhibited significant antimycobacterial activity against *M. tuberculosis* H37Ra (MIC 1.3 μM) [49].

From the isolated compounds **69**, **72**–**75**, **80**–**81**, **83**, **85**, and **87** were tested for cytotoxicity against three cancer cell lines (NCIH187, MCF-7 and KB) and nonmalignant Vero cells. The antimalarial activity against *Plasmodium falciparum* K1, and antitubercular activity against *M. tuberculosis* H37Ra were also carried out on the same compounds [49]. Compounds **69**, **72**–**75**, **80**, **85**, and **87** demonstrated no activity or weak activities in these three assays. However, in contrast, ganoderic acid T (**83**) and its C-3 epimer compound (**81**) exhibited all of the tested activities. *Ganoderma orbiforme* contains the *Ganoderma lanostanoids* comprising the rearranged analog, ganorbiformin A (**69**). Since compound **81** indicated potent antitubercular activity (MIC 1.3 μM) and demonstrated weak cytotoxicity to noncancerous Vero cells (IC_{50} 16 μM) [49].

Wu et al. [60] purchased ganoderic acid (**88**), known as *G. lucidum* triterpenoid, from ChromaDex Inc. Chemicals (Santa Ana, CA), and **88** was tested against cell proliferation and colony formation in MCF-7 human breast cancer cells. Compound **88** showed activity in both assays. Compound **88**-mediated G1 cell cycle arrests both concentration and time dependently. It also significantly decreased the protein level of CDK6, CDK2, p-Rb, cycle D1, and c-Myc in MCF-7 cells [60].

Eighteen lanostane triterpenes; namely, methyl ganoderate A acetonide (**89**) and *n*-butyl ganoderate H (**90**), methyl ganoderate A (**91**), ganoderic acid B (**92**), ganoderic acid E (**93**), ganolucidic acid A (**94**), ganodermadiol (**95**), ganoderic acid Y (**96**), ganoderiol F (**97**), lucidumol B (**98**), ganodermanondiol (99), ganodermanontriol (**100**), lucidadiol (**101**), lucidenic acid N (**102**), lucidenic acid A (**103**), methyl lucidenate E_2 (**104**), *n*-butyl lucidenate N (**105**), and *n*-butyl lucidenate A (**106**) were obtained from *G. lucidum* [50]. The isolated compounds **89**–**106** were tested for their anticholinesterase activity by the spectrophotometric Ellman method with berberine chloride as the reference standard. All the tested compounds gave moderate inhibition to acetylcholinesterase

enzyme, with IC_{50} values ranging from 9.40 μM to 31.03 μM. The IC_{50} value of berberine chloride used as a positive control was 0.04 ± 0.01 μM. Among the compounds *n*-butyl ganoderate H (**90**), *n*-butyl lucidenate N (**105**), and *n*-butyl lucidenate A (**106**) indicated superior activity against acetylcholinesterase with IC_{50} values of 9.40 ± 0.88, 11.58 ± 0.36, and 12.26 ± 0.68 μM, respectively. On the contrary, all the compounds except lucidadiol (**101**) and lucidenic acid N (**102**) displayed no butyrylcholinesterase inhibitory activity. These results indicate that lanostane triterpenes show potential as drug candidates for acetylcholinesterase inhibition [50].

Previous studies on the same group on *G. lucidum*, called reishi, provided information on 18 triterpenoids. The compounds were elucidated as butyl ganoderate A (**107**), butyl ganoderate B (**108**), *n*-butyl lucidenate N (**105**), and *n*-butyl lucidenate A (**106**), ganoderic acid A (**109**), methyl ganoderate A (**91**), ganoderic acid B (**92**), methyl ganoderate B (**110**), methyl ganoderate D (**111**), ganoderic acid E (**93**), methyl ganoderate E (**112**), ganolucidic acid A (**94**), methyl ganoderate H (**113**), lucidenic acid N (**102**), methyl lucidenate A (**114**), methyl lucidenate P (**115**), methyl lucidenate E_2 (**104**), and methyl lucidenate F (**116**) [50]. The inhibitory effect of isolated compounds (**91**–**92**, **94**, **102**, **104**, **106**–**116**) was studied on adipocyte differentiation in 3T3-L1 cells [61]. When compared with the nontreated control, nearly all of the compounds at 40 μg/mL concentration showed inhibition between 22% and 56% of lipid deposit, indicative of 3T3-L1 cell differentiation. Butyl lucidenate N (**105**) indicated the highest inhibition of lipid droplet formation (56%) among the others. The treatment of *n*-butyl lucidenate N (**105**) in a dose-dependent manner reduced the accumulation of lipid droplets in the cells. Compound **105** also suppressed GPDH activity effectively to lipid accumulation. As known, in the triglyceride synthesis pathway, the cytosolic enzyme GPDH plays a central role [61].

A previous study on *G. lucidum* provided five triterpenoids (**91**, **103**, **109**, **117**–**118**); namely, ganoderic acid Sz (**117**), ganoderic acid C1 (**118**), ganoderic acid A (**109**), methyl ganoderate A (**91**), and lucidenic acid A (**103**) [61]. The anticomplement activity of these triterpenoids (**91**, **103**, **109**, **117**–**118**) against the classical pathway of the complement system was examined and compound 116 was found to be potent anticomplement indicating IC_{50} value of 44.6 μM, whereas compounds **91**, **103**, **109**, **118** were determined to be inactive [62].

Lucidenic acid N (**102**), lucidenic acid A (**103**), lucidenic acid B (**119**), and lucidenic acid C (**120**) were isolated from *G. lucidum* strain (YK-02) [62]. These lucidenic acids (**102**–**103**, **119**–**120**) were investigated for their anti-invasive effect against HepG2 cells. The interesting study was done based on the treatment of the lucidenic acids (almost 50 μM) with 200 nM phorbol 12-myristate 13-acetate (PMA). After 24 h incubation of this mixture, the anti-invasive effect against HepG2 cells was observed. According to Weng et al. [63], the lucidenic acids (**102**–**103**, **119**–**120**) are the anti-invasive active compounds on hepatoma cells [63].

Lipophilic extract of *G. lucidum* had also yielded another two new lanostanoids; namely, 7-oxo-ganoderic acid Z (**121**), 15-hydroxy-ganoderic acid S (**122**),

together with ganoderiol F (**97**) and ganodermic acid Q (**123**) [63]. The isolates **121** and **122** showed inhibition activity against the HMG-CoA reductase with IC_{50} values of 22.3 and 21.7 μM, respectively, and against acyl CoA acyltransferase with IC_{50} values of 5.5 and 47.3 μM, respectively. In fact, up to date several oxygenated lanostane-type triterpenoids have been obtained from *G. lucidum* and nearly all showed HMG-CoA reductase-inhibiting activity [64].

The mushroom *G. lucidum* is rich of lanostane-type triterpenoids. The fruiting bodies of *G. lucidum* was also studied by Gao et al. [65] and 10 lanostante-type triterpenes including 3 new lanostante aldehydes; namely lucialdehydes A (**124**), B (**125**), and C (**126**), along with ganodermanonol (**127**), ganodermadiol (**95**), ganodermanondiol (**99**), ganodermanontriol (**100**), ganoderic acid A (**109**), ganoderic acid B (**92**), and ganoderic acid C1 (**118**) have been isolated [65].

The cytotoxicity of isolated triterpenoids were tested against Lewis lung carcinoma (LLC), T-47D (high PR expression of breast cancer cell lines), Meth-A (murine sarcoma), and Sarcoma 180, tumor cell lines. Among the compounds, lucialdehydes B (**125**), C (**126**), ganodermanonol (**127**) and ganodermanondiol (**99**) were found to exhibit cytotoxic effects on T-47D, LLC, Meth-A, and Sarcoma 180 tumor cells, of particular importance is lucialdehyde C (**126**) which demonstrated the highest cytotoxicity against tested tumor cell lines with ED_{50} values of 10.7, 4.7, 3.8, and 7.1 μg/mL, respectively. Ganodermanonol (**127**), however, only exhibited good cytotoxicity against Meth-A cells [65].

Compounds (**128**–**129**) capable of inhibiting selective eukaryotic DNA polymerase activity were isolated from the basidiomycete of *G. lucidum*. The terpenoids were identified as lucidenic acid O (**128**) as well as lucidenic lactone (**129**) [66]. These two terpenoids (**128**–**129**) inhibited the activities of calf DNA polymerase A and rat DNA polymerase B, and also these triterpenoids (**128**–**129**) restrained the human immunodeficiency virus type 1 reverse transcriptase [66].

Ganoderma amboinense is another species of *Ganoderma* Genus. From the basidiomycetous of *G. amboinense*, 16 lanostane triterpenes, 1 of which is new and was identified and elucidated as ganodermacetal (**130**) having an uncommon 7,15-acetonide unit, were isolated. In addition to extensive spectroscopic analyses, chemical transformation reactions were also performed to identify the compounds [67]. Other triterpenoids isolated together were identified as follows: methyl ganoderate C (**131**), ganoderic acid C (**132**), ganoderic acid F (**133**), lanosta-7,9(11), 24-trien-3β,15α, 22-triacetoxy-26-oic acid (**134**), ganodermatriol (**135**), ganoderic acid D (**136**), ganoderic acid DM (**137**), 15-hydroxy-ganoderic acid S (**122**), methyl ganoderate E (**112**), ganodermanontriol (**100**), ganoderiol F (**97**), methyl ganoderate B (**110**), ganoderic acid P (**138**), 3β,15α,22-trihydroxylanosta-7,9(11),24-trien-26-oic acid (**139**), and ganoderic acid H (**140**), ganoderic acid N (**141**) [67].

The *in vitro* toxicity of compounds **97**, **100**, **130**, **131**, **133**, and **134** were studied against brine shrimps (*Artemia salina*) larvae by comparing with chaetomugilin A as a positive reference compound isolated from an endophyte

Chaetomium globosum. The four compounds **100**, **97**, **130**, and **131**, were found to present remarkable toxicity with lethality rates of 91.5%, 81.8%, 70.3%, and 75.8%, respectively. At the same conditions, the positive reference chaetomugilin A demonstrated 78.3% lethality rate [67].

Ganoderic acid X (**142**), which has been reported to inhibit various cancer cell lines growth, is a lanastanoid-type triterpenoid and has been isolated with other anticancer triterpenoids from *G. amboinense* by Li et al. [57]. Some of these triterpenoids also showed inhibitory activity against topoisomerases I and IIα *in vitro* [57].

When ganoderic acid X (**142**) was treated with human hepatoma cell lines (HuH-7) it was noted that it inhibited the DNA synthesis. It also activated apoptosis of hepatoma cells and increased activity of mitogen-activated protein kinases; namely, the extracellular signal-regulated kinase (ERK) and the c-Jun *N*-terminal/stress-activated protein kinase (JNK) [57]. Since ganoderic acid X inhibited both the topoisomerase I and IIα as well as sensitized the cancer cells and thereby promoted apoptosis. Compound (**142**) shows potential as a therapeutic agent for cancer therapy [57].

Five lanostane-type triterpenes, colossolactone V (**143**), colossolactone VI (**144**), colossolactone VII (**145**), and colossolactone VIII (**146**), colossolactone E (**147**) and colossolactone A (**150**) were isolated from *Ganoderma colossum*. HIV-1 protease inhibitory activity of compounds **143**–**147** together with schisanlactone A (**148**) and colossolactone G (**149**) which were previously isolated compounds by Salah et al. [68] from the same mushroom were also evaluated. The compounds schisanlactone A (**148**), colossolactone E (**147**), colossolactone V (**143**), and colossolactone VII (**145**) found to be active against HIV-1 protease with IC_{50} values of 5.0, 8.0, 9.0, and 13.8 μg/mL, respectively [69].

Seven triterpenes (**151**–**157**) were isolated from *Ganoderma pfeifferi*, and three of these were the novel triterpenes (lucialdehyde D, **151**; ganoderone A, **152**; and ganoderone C, **153**). Previously isolated triterpenes examined include: lucialdehyde B (**154**), ganoderol A (**155**), ganoderal A (ganoderol B) (**156**), applanoxidic acid G (**157**), and ganoderone B (lucidadiol) (**104**) [70]. Influenza A and herpes simplex type I (HSV) viruses were used to determine the antiviral activity of the triterpenoids (**152**–**156**). Remarkable activity was observed for lucialdehyde B (**154**) against herpes simplex virus (IC_{50}: 0.075 μg/mL), followed by ganoderal A (ganoderol B) (**156**), and ganoderone A (**152**) with IC_{50} values of 0.03 and 0.3 μg/mL, respectively. In contrast, applanoxidic acid G (**157**) and ganoderone B (lucidadiol) (**104**) did not possess antiviral activity. Likewise, ganoderone A (**152**), lucialdehyde B (**154**), ganoderol A (**155**), and applanoxidic acid G (**157**) demonstrated no antimicrobial activity against tested microorganisms; namely, *Escherichia coli*, *Staphylococcus aureus*, and *Candida maltosa* [70].

The same group [51] studied the European Basidiomycete *G. pfeifferi* and reported that this mushroom possessed the triterpenes ganodermadiol (**98**), lucidadiol (**104**), and applanoxidic acid G (**157**). Antiviral activity of the compounds

were also tested against influenza virus type A and HSV type 1. All compounds demonstrated activity against both virus types [51].

Another *Ganoderma* species *Ganoderma fornicatum* were found to contain two 3,4-seco-25,26,27-trinorlanostane-type triterpeniod compounds; namely, fornicatin A (**158**) and fornicatin B (**159**) together with methyl lucidenate H (**160**) [71]. The inhibitory effects of **158** and **160** on PAF-induced (Platelet-activating factor), ADP-induced (Adenosine diphosphate), and AA-induced (Arachidonic acid) rabbit platelet aggregation were investigated. The results indicated that both compounds **158** and **159** were potent inhibitors on PAF-induced platelet aggregation. Compound **158** also demonstrated mild inhibitory activity against ADP-induced platelet aggregation [71].

The study on *Ganoderma concinna* by *Gonzalez et al*. [72] yielded 12 compounds, of which 3 were new and identified as 5α-lanosta-7,9(11),24-triene-3β-hydroxy-26-al (**161**), 5α-lanosta-7,9(11),24-triene-15α-26-dihydroxy-3-one (**162**) and 8α,9α-epoxy-4,4,14α-trimethyl-3,7,11,15,20-pentaoxo-5α-pregrane (**163**). The nine previously identified triterpenoids were namely, ganoderal A^3 (**164**), ganodermenonol (**165**), ganodermadiol (**98**), ganoderic acid Y (**99**), ganoderiol F (**100**), ganodermatriol (**138**), ganodermanontriol (**103**), ganoderiol A (**166**), and ganoderiol B (**167**) [72]. *Gonzalez et al*. [72] tested biological activity of only the three new compounds and the results indicated that **161**, **162**, and **163** induced apoptosis in human promyelocytic leukemia HL-60 cells [72].

From the *G. tsugae* Murr. (Polyporaceae) three new compounds were identified, namely, tsugaric acid A (**171**), B (**172**), and tsugarioside A (**173**). Four known compounds, 3β-hydroxy-5α-lanosta-8, 24-dien-21-oic acid (**174**), 3-oxo-5α-lanosta-8, 24-dien-21-oic acid (**175**) were also identified from this source [73,74]. Continuous studies with the same mushroom of same group gave other lanostanoids; namely, tsugaric acid C (**168**), tsugariosides B (**169**), and C (**170**) [74]. The cytotoxic activity of the compounds (**170**–**175**) isolated in two studies were investigated against CaSKi, HT-3, PLC/PRF/5, SiHa, T-24, and 212 cell lines. The actinomycin D and cisplatin were used as positive controls. Compound **173** indicated noteworthy activity against T-24 cell line, while **171** and **174**, however, exhibited noteworthy activity against HT-3, T-24, and CaSKi cell lines [75].

Russula lepida and *Russula amarissima* extracts contained nine sesquiterpenoids **176**–**184**, including nordosinanes (**177** and **178**) and aristolanes-type sesquiterpenoids (**176**, **179**–**184**), two triterpenoids (**186**–**187**), and three seco-ocucurbitane triterpene acid (**188**–**190**) [76]. The names of the isolated compounds are aristola-1(10), 8-dien (**176**), rulepidiene B (**177**), rulepidanol (**178**), (+)-aristolone (**179**), rulepidol (**180**), rulepidadiol B (**181**), rulepidadiol C (**182**), rulepidatriol B (**183**), rulepidatriol C (**184**), rulepidatriol B (**185**), lepida acid A (**186**), rosacea acid B (**187**), 3,4−secocucurbita−4,24*E*−diene−3,26-dioic acid (**188**), cucurbitane hydroxyacid (**189**), and 3,4-secocucurbita-4,24*E*-diene-3,26,29-trioic acid (**190**) [76].

Compounds **181**, **182**, **188–190** were tested for their inhibitory activity on the proliferation of WISH, CAKI 1 and A549 cell lines. A moderate activity was found only for the cucurbitane hydroxyacid (**189**). The IC_{50} values were 72, 85.4, and 90.3 μM, respectively. The situated bioactivity of secocucurbitane **189** indicates that **189** may lead to further specific assays to be tested [76].

A novel cyathane diterpenoid, identified as scabronine M (**191**), along with sarcodonin I (**192**) was isolated from the mushroom *Sarcodon scabrosus* [77]. The isolated compounds (**191–192**) were tested for their effects of nerve growth factor (NGF)-induced neurite outgrowth on PC12 (rat pheochromocytoma) cell line. Among them scabronine M (**191**) exhibited inhibitory activity [77].

Studying the same Chinese mushroom species, *S. scabrosus*, also gave cyathane-type diterpenoids. Their structures were elucidated as scabronines K (**193**) and L (**194**), sarcodonin G (**195**), sarcodonin A (**197**), sarcodonin M (**198**), and scabronine H (**196**). In addition, the synthetic compound 19-*O*-acetylsarcodonin G (**199**) was obtained by acetylation of **195** [78]. The isolated compounds (**193–199**) were screened for their NGF-induced neurite outgrowth on PC12 cell line at 25 μM. After 24 h treatment, only sarcodonin G (**195**) and sarcodonin A (**197**) showed neurite outgrowth promoting activity in the presence of 20 ng/mL NGF [78].

Sarcodonin G (**195**), a cyathane-type diterpene from the mushroom *S. scabrosus* [79–80], was found to suppress inflammation induced by TPA which is a tumor promoter. It also inhibited proliferation of HeLa cells, when compared with among other tested cyathane diterpenes (**200–202**). After 2 days of cultured of cells with sarcodonin G (**195**) showed an IC_{50} of 20 μM against HeLa cells in dose-dependent generation [80].

In addition, Western Blot analysis indicated that, **195**-treated HeLa cells were responsible to activate of caspase-3 and caspase-9 as well as to increase of Bax/Bcl-2 ratios. Thus, caspase inhibitor, Z-VAD-FMK, lessen the antiproliferative effect on HeLa cells treated by compound **195** [79]. Compound **195** also exhibited antiproliferative activity against HOC-21, HEC-1, U251-SP, MM-1CB, and HMV-1 human cancer cell lines [80].

Sarcodonin G (**195**) was also investigated for its antibacterial activity against *Bacillus subtilis* and *S. aureus*, and a weak antibacterial activity was observed [52].

Literature shows that *S. scabrosus* mushroom is rich in cyathane diterpenes. Kamo et al. [53] isolated four natural cyathane diterpenes (**203–204**, **208–209**); namely, sarcodonin A (**203**), neosarcodonin O (**204**), allocyathin B_2 (**208**), and neosarcodonin A (**209**) from the mushroom *S. scabrosus* and the anti-inflammatory activity of them were also evaluated by the mouse ear inflammatory test. The authors also prepared derivatives (**205–207** and **210–214**) from the compound **203** [52]. The derivatives (**205–207** and **210–214**) exhibited anti-inflammatory activity between 43% and 78%. At the same conditions, natural diterpenoid **203**, however, demonstrated 75% activity. Other natural diterpenoids **204** and **208** indicated weak activities which are lower than 40% [53].

The compounds sarcodonin A (**203**) and allocyathin B_2 (**208**) were also investigated for their antibacterial activity against *S. aureus* and *B. subtilis*, and weak antibacterial activity was observed [52]. *Shibata et al*. [52] isolated sarcodonin M (**198**) from the mushroom *S. scabrosus* and also prepared a derivative, sarcodonin L, (**215**) by oxidation by using Jones reagent. Both **198** and **215** exhibited good antibacterial activity [52]. *Shi et al*. [81] also isolated secoscabronine M (**216**) which is a hemiacetal cyathane diterpenoid, and scabronine A (**217**) from *S. scabrosus* (Fr.) Karst [81].

The mushroom *S. scabrosus* studied by *Ma et al*. [82] were reported to contain the cyathane-type diterpenoids; namely, scabronine G (**218**) and scabronine H (**219**) were isolated together with four known compounds, allocyathin B_2 (**208**), sarcodonin A (**203**), sarcodonin G (**197**), and scabronine F (**220**) [82].

The other source of diterpenes is *Sarcodon cyrneus* mushroom from which four cyathane diterpenes, cyrneine A (**221**), B (**222**), C (**224**), and D (**225**), were isolated, along with glaucopine C (**223**). Effects of the cyrneines (**221**–**222**, **224**–**225**) and glaucopine C (**223**) on the NGF gene expression in human astrocytoma (1321N1) cells and on neurite outgrowth on pheochromocytoma (PC12) cells were investigated [83]. While, cyrneine B (**222**) ingenerated the strongest NGF expression (7.3-fold at 200 μM concentration), cyrneine A (**221**), cyrneine C (**224**), cyrneine D (**225**), and glaucopine C (**223**) inspired NGF expression level by 3.8-, 2.7-, 1.3-, and 3.5-fold, respectively. These results mean that cyrneine A (**221**), cyrneine B (**222**), and glaucopine C (**223**) induce neurite outgrowth on PC12 cells and also they stimulated NGF gene expression in 1321N1 cells [84]. The same group also investigated the effects on morphology of cyrneine A (**221**) on PC12 cell lines [83]. In a recent study, Marcotullio et al. isolated cyrneine E (**226**), as new cyathane diterpene from *S. cyrneus* [84].

Antrodia camphorata has been used as a chemopreventive agent in Asian folk medicine in Taiwan. *Du et al*. [85] studied the triterpenoid-rich fraction of the mushroom. By using HPLC, the compounds were characterized. It was found that the ethyl acetate extract caused DNA damage in leukemia HL 60 cells as well as apoptotic effect. From the active extract five triterpenoids, antcin K (**227**), antcin C (**228**), zhankuic acid C (**229**), zhankuic acid A (**230**), and dehydroeburicoic acid (**231**) were elucidated. The cytotoxic activity of these triterpenoids (**227**–**231**) was also investigated by the same group. The most potent cytotoxic component was found to be dehydroeburicoic acid (**231**). In HL 60 cells, dehydroeburicoic acid (**231**) induced G2/M phase arrest dose dependently. Compound (**231**) was also showed the activities which its extract also demonstrated. Besides those activities, **231** also inhibited topoisomerase II [85].

Treatment of **231** in HL 60 cells xenograft animal model resulted in a noticeable lessening of tumor weight and size. The xenograft animal model by performed *Du et al*. [85] showed that the antitumor effect of (**231**) was achieved by reducing tumor weight and size [85].

Inonotus obliquus is an important mushroom species, which belongs to the family *Hymenochaetaceae*. Six triterpenoids (**232**, **233**, **234**–**237**) were isolated

from this mushroom collected from Alaska. Their structures were elucidated as lanosterol (**232**), 3β-hydroxylanosta-8,24-dien-21-al (**233**), inotodiol (**234**), trametenolic acid (**235**), lanosta-8,25 dien-3,22,24-triol (**236**), and betulin (**237**) [86].

From *Spongiporus leucomallellus* four triterpenoids, namely, lanosta-8(9), 24(28)-diene-3*β*-ol (**238**), eburicoic acid (**239**), spongiporic acid A (**240**), and spongiporic acid B (**241**) were isolated and identified. Antibacterial and antifungal activities of spongiporic acid A (**240**) was assessed [86]. It was determined that although, it showed weak activity at 10 μg/disc dose against *Proteus vulgaris*, *B. subtilis* and *Bacillus brevis*, it exhibited activity at 50 μg/disc against *P. vulgaris* and clear inhibition zones against *B. subtilis*, *B. brevis*, and the fungus *Paecilomyces variotii* [87].

The mushroom *Leucopaxillus gentianeus* was studied again by *Clericuzio et al*. [88] and it was found to contain four cucurbitane triterpenes; namely, cucurbitacin B (**242**), its fatty acid derivatives (**243**), cucurbitacin D (**246**) and 16-deoxycucurbitacin B (**247**) as well as together with other leucopaxillone triterpenes; namely, leucopaxillone A (**244**) and B (**245**) and 18-deoxyleucopaxillone A (**248**) [88].

Although compounds **242**, **244**–**246**, were isolated from both cultivated mycelia and the fruiting body of the same mushroom; alternatively, compound **247** and a fatty acid ester mixtures of **243** were absent in the mycelia. Likely, 18-deoxyleucopaxillone A (**248**), was isolated from the mycelia, though it was not detected in the mushroom bodies. Triterpenes (**242**–**248**) were tested for their antiproliferative activity against the NCI-H460 human tumor cell line, compared with that of the antitumor agent topotecan, which is known as topoisomerase I inhibitor [88]. The IC_{50} value of cucurbitacin B (**242**), the most active one among the studied triterpenoids (**242**–**248**), was found to be close topotecan's. When the cucurbitacin B (**242**) (IC_{50}: 0.011 μg/mL) and cucurbitacin D (**246**) (IC_{50}: 0.12 μg/mL) compared, role of acetylation at the OH-25 position at C-25 became important. Acetylation of group at C-25 position enhances the bioactivity as seen from the activities of **242** and **246**. Among the leucopaxillones (**244**, **245**, and **248**) leucopaxillone B (**245**) was moderately active. As for 16-deoxycucurbitacin B (**247**), it exhibited lesser active than **242** [88].

From the mushroom *L. gentianeus* (syn. *L. amarus*), bioactive triterpene cucurbitacin B (**249**) and its esters 16-oleyl (**250a**), 16-linoleyl (**250b**), and 16-palmityl (**250c**), together with two novel cucurbitane triterpenoids; namely, leucopaxillones A (**244**) and B (**245**) have been obtained [89]. Cucurbitacin B (**249**) causes a bitter taste in the mushroom; however, its amount is very less. Possibly, during biosynthesis it esterifies to its esters (**250a-c**), which are tasteless compounds [89]. Among cucurbitacins, leucopaxillones A (**244**) and B (**245**) showed oxygenation pattern. The MTT assay was used to assess the growth inhibitory effects of compounds **244**–**245**, **249**, **250a**–**250c** on proliferation of four different human tumor cell lines such as, MCF-7, HepG2, kidney carcinoma CAKI-1 and A549. Cucurbitacin B (**249**) indicated the highest

activity against all tested cell lines. Leucopaxillone A (**244**), however, showed a specific growth inhibitory activity on the MCF-7 cell line [89].

Another lanostane triterpenoid, 24(*E*)-3*β*-hydroxylanosta-8,24-dien-26-al-21-oic acid (**251**), cytotoxic to several cancer cell lines, was isolated from the mushroom *Hebeloma versipelle*. Cytotoxic activity of compound **251** against HL60, Bel-7402, SGC-7901, and A549 tumor cell lines was investigated and the IC_{50} values were found to be 11.2, 20.9, 22.6, and 25.0 μg/mL, respectively. However, compound **251** is quite rare in higher mushrooms [90]. However, the activity might be related to the *α*,*β*-unsaturated aldehyde group as a side chain of lanostan type triterpenoids [90].

Three triterpene curustulinol esters, saponaceols A (**252**), B (**253**), and C (**254**), were isolated from *Tricholoma saponaceum* [91]. The inhibitory effect of saponaceol A (**252**) on cell growth was evaluated in HL-60 human leukemia cells and it was found moderately active with an IC_{50} of 8.9 μM [91].

Eight lanostanoids, elfvingic acid A (**255**), elfvingic acid B (**256**), elfvingic acid C (**257**), elfvingic acid D (**258**), elfvingic acid E (**259**), elfvingic acid F (**260**), elfvingic acid G (**261**), elfvingic acid H (**262**), and the methyl ester of elfvingic acid H (**263**) were isolated from *Elfvingia applanata* [92]. Compounds **255**–**262** and **263** were assayed for their cytotoxicities against Ehlrich cells and Kato III. While, only compound **263** exhibited cytotoxicity (IC_{50}: 1.1 μg/mL for both Ehlrich cells and Kato III cells), in the same conditions the positive control, hinokitiol, indicated 0.6 μg/mL [92].

By using LC-NMR and LC-MS techniques, fasciculol-type triterpene compounds; namely, fasciculic acids D (**264**), E (**265**), and F (**266**) were detected and then isolated from the mushroom *Hypholoma fasciculare* [93]. As for the biological activities of the compounds (**264**–**266**), triterpenoids **265** and **266** inhibited moderately 3*α*-hydroxysteroid dehydrogenase (IC_{50}: 14.0 μM). However, **264**, **265**, and **266** showed weak antimicrobial activity on some gram-positive bacteria such as *B. subtilis* ATCC 6633 (MIC > 500 μg/mL). Investigation of fasciculol and fasciculic acid productions in other two *Hypholoma* species (*H. sublateritium* and *H. capnoides*) resulted that while, *H. sublateritium* can produce fasciculol D (**264**) and fasciculol F (**266**), *H. capnoides* was incapable of producing fasciculols [93].

From the solid culture of *Cyathus africanus*, the medicinal fungus, eight cyathane-type diterpenes; namely, cyathins D–H (**267**–**271**), neosarcodonin O (**204**), cyathatriol (**272**), and 11-*O*-acetylcyathatriol (**273**) were isolated. In addition, 11,15-*O*,*O*-diacetylcyathatriol (**274**) and 11,14,15-*O*,*O*-triacetylcyathatriol (**275**) were prepared by the acetylation of compound **273** [94]. All of the isolated compounds (**204**, **267**–**275**) were tested for their NO production inhibitory activities from lipopolysaccharide-activated macrophage RAW 264.7 and for their antitumor activity against HeLa and K562 (human erythromyeloblastoid leukemia cell line) cell lines [94]. Compounds **204**, **269**, **271**, **273**, and **274** showed effective inhibitory activity against nitric oxide production in lipopolysaccaride-activated macrophages with IC_{50} values of 2.57, 1.45, 12.0,

10.73, and 9.45 μM, respectively. The positive control hydrocortisone indicated 53.78 μM IC_{50} value at the same conditions. Compounds **204** and **273** showed strong cytotoxic activity against K562 and Hela lines indicating the IC_{50} values less than 10 μM. Compound **271**, however, showed only mild cytotoxic activity on both cell lines with IC_{50} values of 23.72 and 39.46 μM, respectively [94]. A cyathane-xyloside type diterpene erinacine A (**276**) having effective stimulating activity on the NGF synthesis was isolated from *Hericium erinaceum* which is called edible medicinal lion's mane mushroom. Erinacine A (**276**) is known as antibacterial agent particularly against methicillin-resistant *S. aureus* [95, 96].

Tricholomalides A (**277**), B (**278**), and C (**279**), neurotrophic diterpenes, were isolated for the first time from the mushroom *Tricholoma* sp. The trichoaurantianolides B (**280**) and C (**281**) which are the metabolites of *T. aurantium*, were also isolated by *Tsukamoto et al*. [97]. To determine potential bioactivity, the isolated compounds were subjected to neurite outgrowth assay and cytotoxicity test against HeLa cell lines. Tricholomalides A (**277**), B (**278**), and C (**279**) induced neurite outgrowth in rat PC-12 cells at 100 μM concentrations [97]. Studies on the chemistry, isolation, and elucidation of diterpenoid compounds have been performed and are noted in the literature [97]. However, the biological activity of these compounds has not been assessed [97]. Fang et al. [98] isolated cyathane diterpenoids from the fruiting bodies of basidiomycetes of *Phellodon niger*. They were elucidated as nigernin A (**282**) and B (**283**) [98].

The Steroids

Steroid is a type of organic compound that contains a characteristic arrangement of four cycloalkane rings that are joined to each other [99]. They are lipophilic compounds that play a significant role in biological systems [100]. The natural steroids, cholesterol and cholic acid, are the firstly known steroids [101]. Also, hundreds of different steroids such as bile acids, phytosterols, and steroidal hormones have been isolated from animals, fungi, and plants [99, 100]. Steroids are responsible for important biological functions in the cells. For example, androstane, pregnane, and estrane exhibit various hormonal activities, the bile acids are essential for lipid digestion and absorption, the cardiac aglycones are used for heart failure treatment [102], and the dexamethasone are used as an anti-inflammatory drug [103]. Several steroids have been isolated from different mushrooms that have anticomplement [62], antimicrobial [70], cytotoxic [104], antitubercular [105], and antiviral [106] activities. The biologically active steroids and their activity values were given in Table 3, and the structures in Figure 1.

The methanol extract of *Sarcodon joedes* was reported to contain nine compounds, including four steroids (22*E*, 24*R*)-6*β*-methoxyergosta-7,22-diene-3*β*,5*α*-diol (**284**), ergosta-7,22-dien-3*β*-ol (**285**), ergesterol peroxide (**286**) and (22*E*,24*R*)-3*β*-hydroxyergosta-5,22-dien-7-one (**287**) [104]. The cytotoxic activity of isolated compounds was also tested by the same group. The compound ergosterol (**288**) had already isolated from the *Paxillus involutus*

TABLE 3 The Biologically Active Steroids and Their Activity Values

Mushroom Species	Isolated Steroids	Bioactivities Tested with Activity Values	References
Sarcodon joedes	**284**	*Cytotoxic activity*: IC_{50}: 50.23 μM against the U87 IC_{50}: 32.69 μM against LO2	[104]
Hypsizigus marmoreus	**286**	*Antitubercular activity*: MIC: 1 μg/mL against $H_{37}Rv$	[105]
Ganoderma lucidum		*Anticomplement activity*: IC_{50}:126.8 μM	[62]
H. marmoreus	**288**	*Antitubercular activity*: MIC > 128 μg/mL against $H_{37}Rv$	[105]
G. lucidum		*Anticomplement activity*: IC_{50}: 52.0 μM	[62]
Fomes fomentarius	**294**	*Cytotoxic activity*: IC_{50}: 180 μM against the HCT116	[113]
F. fomentarius	**296**	*Cytotoxic activity*: IC_{50}: 160 μM against the HCT116	[113]
F. fomentarius	**301**	*Cytotoxic activity*: IC_{50}: 125 μM against the HCT116	[113]
F. fomentarius	**302**	*Cytotoxic activity*: IC_{50}: 1.3 μM against the A549 IC_{50}: 2.1 μM against the MCF-7 IC_{50}: 17.5 μM against the NUGC-3	[113]
Tomophagus cattienensis	**324**	*Cytotoxic activity*: IC_{50}: 91.2 μM against KB cells	[120]
T. cattienensis	**326**	*Cytotoxic activity*: IC_{50}: 63.3 μM against KB cells	[120]

mushroom [107]. All the compounds **284**–**287** were examined for their inhibitory activity against human glioma U87 and steatotic hepatocytes LO2 cell lines using the MTT assay [108]. Compound **284** showed significant inhibitory cell growth activity against the U87 and LO2 cell lines with IC_{50} values of 50.23 and 32.69 μM, respectively. The other compounds were found to be inactive [104]. Compounds **284**–**286** are found in terrestrial fungi [109].

Akihisa et al. [105] and *Lee et al.* [108] reported that **284** is a potent cytotoxic agent against some cancer cell lines such as murine L1210 leukemia cells, mouse LLC and human K562 cells [105,108]. According to *Itoh et al*. [106] compound **284** causes cell death and morphological change in human lung cancer (LU99) and stomach cancer (KATO III) cells [106]. Ergesterol peroxide (**286**), however, has an ability to inhibit the growth of some cancer cells and to cause apoptosis of human leukemia (HL60) cells [104,110]. Steroid (**286**) also inhibited TPA-induced inflammation and tumor promotion *in vivo*, and suppressed proliferation of mouse and human lymphocytes stimulated with mitogens [104,111]. As known miR-378 and U87 cells have aggressive cancer cell properties and it became chemoresistant. In a recent study by *Wu et al*. [112] ergosterol peroxide (**286**) was isolated from the *G. lucidum* and tested against miR-378 and U87 cells. Ergosterol peroxide (**286**) was found to overcome the chemoresistance of miR-378. Compound (**286**) showed greater activity in causing death of miR-378 cells compared to the GFP cells which are used as a control vector. The study concluded that, if further clinical development of ergosterol peroxide (**286**) is achieved, this compound has the potential to be used to kill off the drug resistance of tumor cells [112].

From the mushroom *S. scabrosus* seven steroids (**286**, **288**–**293**); namely, ergesterol peroxide (**286**), ergosterol (**288**), tuberoside (**289**), 3-*O*-*β*-D-glucopyranosyl-5*α*,8*α*-epidioxyergosta-6,22-diene (**290**), ergosta-7,22-dien-3*β*-ol (**291**), 3*β*,5*α*,9*α*-trihydroxy-6*β*-methoxy-ergosta-7,22-dien (**292**), and ergosta-7,22-dien-3*β*,5*α*,9*α*-trihydroxy-6-one (**293**) were isolated [77]. The compounds (**286**, **288**–**293**) were tested for their inhibitory effects on NGF-mediated neurite outgrowth on PC12 cells, and all of the steroids exhibited weak activity. MTT assay showed that even, at the highest concentration (100 μM), the compound found inactive [77].

From *Fomes fomentarius*, known as the tree mushroom which is a polypore macrofungus, 11 (**286**, **294**–**302**) of which 4 were new ergostane-type sterols, were isolated. Spectroscopic studies revealed that, the structures of sterols **286**, **294**–**302** were identified as ergesterol peroxide (**286**), fomentarol A (**294**), fomentarol B (**295**), fomentarol C (**296**), and fomentarol D (**297**), (22*E*,24*R*)-3*β*-hydroxyergosta-7,22-diene-6-one (**298**), (22*E*,24*R*)-ergosta-7,22-dien-3*β*,5*α*,6*α*-triol (**299**) (22*E*,24*R*)-ergosta-7,9(11),22-trien-3*β*,5*α*,6*β*-triol (**300**), (22*E*,24*R*)-ergosta-7,22-dien-3*β*,5*α*,6*β*,9*α*-tetraol (**301**), 3*β*-*O*-glucopyranosyl-5*α*,6*β*-dihydroxyergosta-7,22-diene (**302**), and cerevisterol (**303**) [113]. Compounds **286**, **294**–**298**, and **303** were screened for their cytotoxic activities against H1299 and HCT116 cancer cell lines by the MTT assay, while the cytotoxic activities of compounds **286**, **294**–**303** against A549, MCF-7, NUGC-3, SHSY-5Y, and SNU739 cancer cells were tested using the CellTiter Glo™ luminescent cell viability assay. In the MTT assay, only compounds **294**, **296** and **301** showed mild cytotoxic effects against colon cancer (HCT116) cells. The positive control in order to compare the activity 5-FU was used. In the CellTiter Glo™ assay, however, only **302** exhibited mild cytotoxic activity against the A549, MCF-7, and the NUGC-3 [113].

Compound **303**, identified as cerevisterol was isolated from the basidiomycete of *G. lucidum* and has been reported to inhibit eukaryotic DNA polymerase activity selectively. This steroid, cerevisterol (**303**) is noted to be cytotoxic and has been shown to supress the activity of DNA polymerase A [66]. From the fruiting bodies of *Stropharia rugosoannulata*, a sterol having an unprecedented ether ring (**304**) and the sterol (**305**) were isolated by *Wu et al*. [114].

Mushroom *S. rugosoannulata* called saketsubatake in Japan, was shown to contain four steroids (**306**–**309**). Compound **306** was isolated previously by *Wu et al*. [115] from the mushroom *S. ragusoannulata* [115]. Compound **307** has also been isolated from other mushroom species; namely, *Amanita panthering*, *Amanita virgineonides*, *Lactarius piperatus*, *Lyophyllum shimeji*, *Tricholoma portentosum*, *Hypsizigus marmoreus*, and *Lentinula edoes* [116]. Compound **308** and **309** are the metabolites of *Tricholoma portensum* and *Grifola frondosa*. *Kwon et al*. [117] also reported moderate cytotoxicity of compound **309** [117]. In addition, *Wu et al*. [115] tested the effect compounds (**307**–**309**) on lettuce plant growth. It was shown that **308** and **309** inhibited the hypocotyl growth of lettuce; however, compound **307** did not demonstrate activity [118].

G. lucidum has been shown to contain four antiaging compounds; namely, ganodermasides A (**310**), B (**311**), C (**312**), and D (**313**) [119]. In a study by *Seo et al*. [62] *G. lucidum* also contains the following steroids (**286**, **288**, **314**); namely, ergesterol peroxide (**286**), ergesterol (**288**), and stella sterol (**314**). Anticomplement activity of **286**, **288**, and **314** were examined and compounds **286** and **288** indicated significant anticomplement activity (IC_{50}: 126.8 and 52.0 μM, respectively), whereas compound **314** was inactive [62].

Three steroids; namely, ergosta-7, 22-diene-3*β*-yl palmitate (**315**), ergosta-7,22-diene-3*β*-yl linoleate (**316**), and ergosta-7,22-diene-3*β*-yl pentadecanoate (**317**) were also obtained from *G. lucidum* [87]. The same researchers also investigated the mushroom *S. leucomallellus* (Murril) ADavid, and ergosterol (**288**), ergosta-7,22-diene-3*β*-ol (**318**), ergesterol peroxide (**286**), and 5,8-epi-dioxy-24-methyl-cholesta-6,9(11),22-triene-3*β*-ol (**319**) were isolated. Compound **319** was tested to determine anticarcinogenic activity against Jurkat T-cell leukemia cells and HeLa-S3 cervix carcinoma; and only mild activity was found [87].

Two steroids (**320**–**321**) were isolated from *G. pfeifferi* and these compounds were identified as ergosta-7,22-dien-3*β*-ol (**320**), and ergosta-7,22-dien-3-one (**321**) [70]. Antiviral activity of the compounds (**320**–**321**) were tested against influenza virus type A and HSV type 1 at noncytotoxic concentrations. According to antiviral activity results, remarkable activity was observed in ergosta-7,22-dien-3*β*-ol (**320**) as it demonstrated strong inhibitory activities against HSV (IC_{50}: 0.03 μg/mL). In contrast, ergosta-7,22-dien-3-one (**321**) did not possess antiviral activity. Antimicrobial activity of both steroids (**320**–**321**) against *E. coli*, *S. aureus*, and *C. maltosa* when tested at 200 μg/disk was not detected [70]. A study on *G. concinna* by *Gonzalez et al*. [72] yielded three known steroids; namely, ergosta-7,22-dien-3-one (**321**), fungisterol (**322**), and ergosterol peroxide (**286**) [72].

The mushroom species naturally grow in Vietnam called *Tomophagus cattienensis* belongs to *Ganodermataceae* family have been recently studied by *Hien et al*. [120]. From the fruiting body of the mushroom four steroids (**323–326**); namely, cattienoids A–C (**323–325**) schisanlactone A (**326**), were isolated [120]. These compounds have an unusual seven-membered lactone ring, derived from lanostane-type triterpenoids.

These four compounds were tested for their cytotoxicity, antimicrobial activity against several bacteria as well as tyrosinase inhibitory activity. It was found that cattienoid B (**324**) and schisanlactone A (**326**) possess cytotoxicity against KB cells (IC_{50}: 91.2 and 63.3 μM, respectively). However, none of them showed antimicrobial activity against *S. aureus*, *B. subtilis*, *Lactobacillus fermentum*, *Salmonella enterica*, *E. coli*, *Pseudomonas aeruginosa*, and *C. albicans* and none of the indicated inhibitory activity against tyrosinase enzyme [120].

From the edible and medicinal mushroom *H. erinaceum*, which is called lion's mane mushroom, ergesterol peroxide (**286**), and ergesterol (**288**) were isolated [95]. Since, ergesterol peroxide indicated cytotoxic activity, in order to find the ergesterol peroxide concentration in other mushroom species, the densitometric method developed by *Krzyczkowski et al*. [121] was applied to *n*-hexane extracts of six mushroom species; namely, *H. erinaceum*, the chicken mushroom (*Laetiporus sulphureus*), and the common morel (*Morchella esculenta*), and the king bolete (*Boletus edulis*), the Jersey cow mushroom (*Suillus bovinus*), and the bay bolete (*Boletus badius*) [121]. The compound **286** was quantified in the mushroom species as 15.98 ± 0.78, 10.07 ± 0.75, 13.37 ± 0.56, 29.32 ± 1.43, 17.27 ± 0.84, and 12.60 ± 0.59 mg per 100 g, respectively. The study showed that *B. edulis* contains compound **286** more than other mushroom species tested [121].

Seven sterols were isolated from *Cantharellus cibarius* mushrooms. Sterols **286**, **288**, **327–331** were identified as ergesterol peroxide (**286**), ergesterol (**288**), tuberoside (**327**), β-sitosterol glucoside (**328**), cerevisterol (**303**) 7-dehydrostigmasterol (**330**) and β-sitosterol (**331**) [122]. β-sitosterol (**331**), also isolated from *I. obliquus* which is an important mushroom species belonging to the Hymenochaetaceae family [86]. The Nuclear Factor kappa B (NF-κB) inhibitory activity of the isolated compounds (**286**, **288**, **327–331**) as well as the extracts of *C. cibarius* was investigated. Among the tested compounds, ergosterol peroxide (**286**), ergosterol (**288**), and cerevisterol (**303**) have the most potent inhibitors of NF-κB activation. Inhibitory activity of **286** was comparable with the activity of rocaglamide, used as the positive control [86].

H. marmoreus yielded seven sterols (**332–338**) which were also assayed for their antitubercular effective against *M. tuberculosis* H37Rv [106]. Six sterols (**333–338**) showed activity, while the steroid **332** was inactive. The seven sterols (**332–338**) were also investigated for their antiviral activity against antigen activation induced Epstein–Barr virus. In Raji cells, this virus is known as the tumor promoter. Compounds **337** and **338** showed effective inhibitory activity and further the best is that these compounds gave no damage to Raji cells [106].

The Phenolic Compounds

Phenolic compounds, a large class of phytochemicals, have interesting biological properties. Anthocyanins, flavonoids, catechins, phenolic acids, secoiridoids, stilbenes, coumarins, and isoflavones are the examples to phenolic compounds. They are available mostly in vegetables, fruits, herbs, grains, and seeds as well as in juices, wines, oils, and etc.

In recent years, there have been many reports in the literature on the role of phenolic compounds. They are the chain breaking antioxidants and prevent the negative effects of oxygen and nitrogen reactive species (ROS/RNS) [123]. In fact, their antioxidant ability is due to their hydroxyl groups [124]. According to the *Tanaka et al*. polyphenolic compounds have inhibitory effects on mutagenesis and carcinogenesis in humans, when up to 1.0 g daily ingested from a diet rich in fruits and vegetables [125]. Furthermore, they can stop or delay some of the human diseases such as cardiovascular, atherosclerosis, cancer, diabetes, Alzheimer's, and Parkinson's diseases [126]. The biologically active phenolic compounds and their activity values were given in Table 4.

Nine anthraquinone-type compounds from the *Cortinarius purpurascens* mushroom were isolated and identified as citreorosein 6,8-dimethyl ether (**339**), physcione (**340**), rufoolivacin (**341**), rufoolivacin C (**342**), rufoolivacin D (**343**), leucorufoolivacin (**344**), verbindung cr11 (**345**), verbindung cr60 (**346**), and 1-hydroxy-3-methyl-2-isopropanyl-6,8-dimethoxyanthraquinone (**347**) [127]. The antioxidant activity of the compounds was also investigated by the same group using the DPPH free radical scavenging activity. Among the tested compounds, **341**–**344** exhibited DPPH free radical scavenging activity with IC_{50} values in the range of 3–8 μg/mL which is comparable with that of TBHQ, used as positive standard antioxidant. Compounds **344** and **341** exhibited very similar activity to the control TBHQ. Moreover, **347** and **344** displayed much higher activity than **345** and **346**. Compound **344**, however, was slightly stronger than **341** since **344** due to having an additional hydroxyl group at C-1. The anthraquinones, **339** and **340** had little or no activity. According to *Bai et al*. [127], these mushroom species are one of the potential sources of antioxidants from natural products [127].

The methanol extract of *S. joedes* also provided four phenolic compounds. The phenolics were identified as 2′,3′-diacetoxy-3,4,5′,6′,4″-pentahydroxy-*p*-terphenyl (**348**), benzoic acid (**349**), methyl *p*-hydroxybenzoate (**350**), and 3,4-dihydroxybenzoic acid (**351**)] [107]. The cytotoxic activity of **348**–**351** was also evaluated [107].

Compound **348** exhibited marked toxicity toward the brine shrimp (Artemia salina) larvae with a LC_{50} value of 63.29 μg/mL. The phenolic metabolites **349**–**351** have been relatively infrequently encountered in mushrooms. The antibacterial properties of the phenolic compound **348** isolated from the fruiting bodies of *Sarcodon leucopus* has been reported previously [111].

TABLE 4 The Biologically Active Phenolics and Their Activity Values

Mushroom Species	Isolated Phenolics	Bioactivities Tested with Activity Values	References
Cortinarius purpurascens	**341**	*DPPH scavenging activity* IC_{50}: 154 ± 2.65 μg/mL	[127]
C. purpurascens	**342**	*DPPH scavenging activity*: IC_{50}: 230 ± 3.58 μg/mL	[127]
C. purpurascens	**343**	*DPPH scavenging activity*: IC_{50}: 4.65 ± 0.02 μg/mL	[127]
C. purpurascens	**344**	*DPPH scavenging activity*: IC_{50}: 8.63 ± 0.10 μg/mL	[127]
C. purpurascens	**345**	*DPPH scavenging activity*: IC_{50}: 7.50 ± 0.12 μg/mL	[127]
C. purpurascens	**346**	*DPPH scavenging activity*: IC_{50}: 3.88 ± 0.65 μg/mL	[127]
C. purpurascens	**347**	*DPPH scavenging activity*: IC_{50}: 36.5 ± 0.60 μg/mL	[127]
Boletus pseudocalopus	**358**	*Cytotoxic activity*: IC_{50}: 8.5 ± 0.6 μg/mL against A549 IC_{50}: 7.3 ± 0.7 μg/mL against B16F1 IC_{50}: 16.9 ± 1.8 μg/mL against SK-Mel-2 *DPPH scavenging activity*: IC_{50}: 72.4 ± 7.8 μg/mL	[131]

TABLE 4 The Biologically Active Phenolics and Their Activity Values—Continued

Mushroom Species	Isolated Phenolics	Bioactivities Tested with Activity Values	References
B. pseudocalopus	**359**	*Cytotoxic activity*: IC_{50}: 9.0 ± 0.9 μg/mL against A549 IC_{50}: 4.3 ± 0.2 μg/mL against B16F1 IC_{50}: 11.9 ± 1.4 μg/mL against SK-Mel-2 *DPPH scavenging activity*: IC_{50}: 63.1 ± 6.2 μg/mL	[131]
B. pseudocalopus	**360**	*Cytotoxic activity*: IC_{50}: 5.0 ± 0.7 μg/mL against A549 IC_{50}: 3.5 ± 0.7 μg/mL against B16F1 IC_{50}: 13.7 ± 1.2 μg/mL against SK-Mel-2 *DPPH scavenging Activity*: IC_{50}: 67.4 ± 4.5 μg/mL	[131]
B. pseudocalopus	**361**	*Cytotoxic activity*: IC_{50}: 10.5 ± 1.3 μg/mL against A549 IC_{50}: 6.1 ± 0.2 μg/mL against B16F1 IC_{50} > 40 μg/mL against SK-Mel-2	[131]
Albatrellus caeruleoporus	**361**	*DPPH scavenging activity*: IC_{50}: 43.8 ± 6.8 μg/mL *NO production inhibitory activity*: IC_{50}: 29.0 μM in RAW 264.7	[135]
B. pseudocalopus	**362**	*Cytotoxic activity*: IC_{50}: 10.5 ± 0.7 μg/mL against A549 IC_{50}: 5.4 ± 0.4 μg/mL against B16F1 IC_{50} > 40 μg/mL against SK-Mel-2 *DPPH scavenging activity*: IC_{50}: 11.1 ± 1.8 μg/mL	[131]
A. caeruleoporus		*NO production inhibitory activity*: IC_{50}: 23.3 μM in RAW 264.7	[135]

Continued

TABLE 4 The Biologically Active Phenolics and Their Activity Values—Continued

Mushroom Species	Isolated Phenolics	Bioactivities Tested with Activity Values	References
B. pseudocalopus	**363**	*Cytotoxic activity*: IC_{50} > 40 μg/mL against A549, B16F1, SK-Mel-2 *DPPH scavenging activity*: IC_{50}: 75.9 ± 3.3 μg/mL	[131]
B. pseudocalopus	**364**	*Cytotoxic activity*: IC_{50}: 11.0 ± 1.1 μg/mL against A549 IC_{50}: 6.0 ± 0.8 μg/mL against B16F1 IC_{50}: 8.0 ± 1.1 μg/mL against SK-Mel-2 *DPPH scavenging activity*: IC_{50}: 67.4 ± 5.2 μg/mL	[131]
B. pseudocalopus	**365**	*Cytotoxic activity*: IC_{50}: 7.5 ± 0.8 μg/mL against A549 IC_{50}: 6.5 ± 1.1 μg/mL against B16F1 IC_{50} > 40 μg/mL against SK-Mel-2 *DPPH scavenging activity*: IC_{50}: 28.6 ± 3.5 μg/mL	[131]
B. pseudocalopus	**366**	*Cytotoxic activity*: IC_{50}: 5.0 ± 0.3 μg/mL against A549 IC_{50}: 4.1 ± 0.4 μg/mL against B16F1 IC_{50} > 40 μg/mL against SK-Mel-2 *DPPH scavenging activity*: IC_{50}: 14.6 ± 2.6 μg/mL	[131]

TABLE 4 The Biologically Active Phenolics and Their Activity Values—Continued

Mushroom Species	Isolated Phenolics	Bioactivities Tested with Activity Values	References
Ganoderma colossum	**367**	*Antiviral activity*: IC_{50}: 7.5 μg/mL against HIV-1 protease	[132]
G. colossum	**368**	*Antiviral activity*: IC_{50}: 1.0 μg/mL against HIV-1 protease	[132]
Ganoderma pfeifferi	**368**	*Antimicrobial activity*: 20 mm zone against *Staphylococcus aureus* ATCC 6538 17 mm zone against *S. aureus* ATCC 25923 18 mm zone against *S. aureus* ATCC 29213 24 mm zone against *S. aureus* SG 511 15 mm zone against *Bacillus subtilis* SBUG 14 26 mm zone against *M. flavus* SBUG 16 No activity against *P. aeruginosa* ATCC 15442 5 mm zone against *E. coli* SBUG 13 15 mm zone against *Proteus mirabilis* SBUG 47 16 mm zone against *S. marcescens* SBUG 9	[134]
G. colossum	**369**	*Cytotoxic activity*: IC_{50}: 15 μg/mL against the Hep-2	[133]
G. fornicatum	**370**	*Cytotoxic activity*: IC_{50}: 30 μg/mL against the Hep-2	[133]
G. fornicatum	**371**	*Cytotoxic activity*: IC_{50}: 23 μg/mL against the Hep-2	[133]

Continued

TABLE 4 The Biologically Active Phenolics and Their Activity Values—Continued

Mushroom Species	Isolated Phenolics	Bioactivities Tested with Activity Values	References
G. pfeifferi	**372**	*Antimicrobial activity*: 19 mm zone against *S. aureus* ATCC 6538 15 mm zone against *S. aureus* ATCC 25923 16 mm zone against *S. aureus* ATCC 29213 24 mm zone against *S. aureus* SG 511 16 mm zone against *B. subtilis* SBUG 14 25 mm zone against *M. flavus* SBUG 16 No activity against *P. aeruginosa* ATCC 15442 4 mm zone against E. coli SBUG 13 15 mm zone against *P. mirabilis* SBUG 47 15 mm zone against S. marcescens SBUG 9	[134]
A. caeruleoporus	**373**	*NO production inhibitory activity*: IC_{50}: 23.4 μM in RAW 264.7	[135]
A. caeruleoporus	**374**	*NO production inhibitory activity*: IC_{50}: 22.9 μM in RAW 264.7 cells	[135]
Tricholoma orirubens	**375**	*Hyaluronan-degradation inhibitory activity*: IC_{50}: 15.0 μM in human skin fibroblasts	[136]
T. orirubens	**376**	*Hyaluronan-degradation inhibitory activity*: IC_{50}: 21.0 μM in human skin fibroblasts	[136]
T. orirubens	**377**	*Hyaluronan-degradation inhibitory activity*: IC_{50}: 57.0 μM in human skin fibroblasts	[136]

Inonutus obliquus collected from Alaska has been found to be a source of two phenolic compounds (**352**–**353**). Their structures were elucidated as 2,5-dihydroxybenzaldehyde (**352**), and 3,4-dihydroxybenzalacetone (**353**) [86]. Among these compounds, **352** was isolated for the first time from this mushroom [86]. *Fang et al.* [98] isolated phellodonin (**354**), the nitrogenous terphenyl derivative of *P. niger* [104].

Inonotus hispidus is a tree mushroom species and grows on *Pistacia atlantica* tree. Three phenolic compounds were isolated from the mushroom and the elucidation of compounds **355**–**357** by spectroscopic techniques yielded a novel natural antioxidant; namely, methyl 5-(3,4-dihydroxyphenyl)-3-hydroxypenta-2,4-dienoate (**355**), along with hispolone (**356**) and hispidin (**357**) [128]. In other studies, hispidin (**357**) was tested for its antiviral and cytotoxicity activities [129,130]. The antioxidant and antiviral properties of hispolon (**356**), however, have been previously reported [129].

By activity-guided fractionation studies, from the methanol extract of *Boletus pseudocalopus*, a Korean wild mushroom, three grifolin derivatives (**358**–**360**) together with four phenolic compounds (**361**–**364**) were isolated [131]. The side chain of the grifolin (**361**) and neogrifolin (**362**) were saturated to obtain **365** and **366** for biological testing.

The cytotoxic effects of the compounds **358**–**364**, and the hydrogenated compounds **365** and **366** were assayed with two different cancer cell lines; namely, A549 and B16F1. Except compound **361**, the IC_{50} of compounds were found to be in the range of 5.0–11.0 and 3.5–7.3 μg/mL, respectively. The results were close to cisplatin used as a positive control [131].

The antiradical activity of compounds **358**–**366** were assessed with the DPPH assay. It was found that hydrogenation was not responsible to the antiradical and cytotoxic activities. Both antiradical and anticancer activities of **365** which were obtained by hydrogenation of **361**, reported as same like those of grifolin (**361**). Similarly, those of **366** were same to neogrifolin's (**362**) [131].

Ganomycin I (**367**), ganomycin B (**368**), and fornicin B (**369**), the farnesyl hydroquinones, were isolated from *G. colossum*. These compounds were also tested for their antiviral activity against HIV-1 protease. The results revealed that ganomycin I (**367**) and ganomycin B (**368**) demonstrated activity against HIV-1 protease presenting IC_{50} of 7.5 and 1.0 μg/mL, respectively [132]. Fornicin B (**369**), however, was evaluated for cytotoxic activity against the Hep-2 cell line and displayed moderate inhibitory activity (IC_{50}:15 μg/mL) [132,133]. Fornicin A (**370**) and fornicin C (**371**) which are prenylated phenolic compounds, have previously been isolated from the *G. fornicatum*. Cytotoxic activity of the compounds **370** and **371** were also screened against the same cell line. Fornicin A (**370**) and fornicin C (**371**) were found to be moderately active against the human larynx carcinoma cancer cell line, indicating IC_{50} of 30 and 23 μg/mL, respectively [133].

Antimicrobial activity against numerous gram-positive and gram-negative bacteria of ganomycin A (**372**) and B (**368**) isolated from *G. pfeifferi* was

tested. The compounds **368** and **372** exhibited antimicrobial activity [134]. In general, ganomycins A (**372**) and B (**368**) indicated moderate growth inhibition of some strains, particularly against gram-positive bacteria with the zones of inhibitions of 15–25 mm. The MIC values of both ganomycins against *Micrococcus flavus* found to be 2.5 μg/mL (at the same conditions the MIC of ampicillin was 0.25 μg/mL). Against *S. aureus*, however, the MIC value was 25 μg/mL measured for both, as well (MIC of ampicillin 0.05 μg/mL). No activity was observed against the yeasts *C. albicans* and *C. maltose* [134].

Grifolinones A (**373**), grifolinones B (**374**), grifolin (**361**), and neogrifolin (**362**), which are farnesyl phenols were purified and elucidated from polar extract of *Albatrellus caeruleoporus* mushroom. Grifolinone B (**374**) was identified as dimer of grifolin (**361**). The isolated farnesyl phenols (**373**–**374**, **361**–**362**) showed activity against nitric oxide (NO) production stimulated by lipopolysaccharide in RAW 264.7 cells (IC_{50}: 23.4, 22.9, 29.0, and 23.3 μM, respectively). The time validation of the test demonstrated that **372**–**373**, **361**–**362** strongly inhibited NO production at 24 h by lipopolysaccharide induced cells (IC_{50}: 23.4, 22.9, 29.0, and 23.3 μM, respectively). When the higher concentrations (more than 50 μM) were used, the cell viability was reduced because of cytotoxicity of **373**–**374**, **361**–**362**. Quang et al. [135] also revealed that significant changes in activity were observed when small structural differences in these compounds occurred [135].

Orirubenone A (**375**), B (**376**), and C (**377**) were isolated from the mushroom *Tricholoma orirubens* and were evaluated for their hyaluronan-degradation inhibitory activity on human skin fibroblasts. Orirubenone A–C (**375**–**377**) exhibited the inhibitory activity indicating IC_{50} values of 15.0, 21.0, and 57.0 μM, respectively [136]. *P. involutus* has been reported to be a source of the phenolic compounds involutin (**378**) and involutone (**379**) [108].

The Alkaloids

The alkaloids are the important class of natural products. Their activity is due to their nitrogen groups. Many natural drugs are alkaloid in structure or have nitrogen in their structures. There are limited numbers of alkaloids that have been isolated from the mushroom species. In this section, the alkaloids isolated from fungi in the last two decades were evaluated herein. The biologically active alkaloids and their activity values were given in Table 5, and the structures in Figure 1.

In a recent study, a norsesquiterpene alkaloid (**380**) was isolated from the fungus *Flammulina velutipes* which is present fermented on rice. **380** was also tested against KB cells *in vitro* by using the MTT assay and it was found to possess moderate cytotoxicity with an IC_{50} value of 16.6 mmol/L [137]. From

TABLE 5 The Biologically Active Alkaloids and Their Activity Values

Mushroom Species	Isolated Alkaloids	Bioactivities Tested with Activity Values	References
Flammulina velutipes	**380**	*Cytotoxic activity*: IC_{50}: 16.6 μmol/L against KB cells	[137]
Ganoderma sinense	**381**	*Activity in protecting H_2O_2-oxidation induced injury*: EC_{50}: 6.2 μmol/L on HUVEC	[138]
Astraeus odoratus	**383**	*Antitubercular activity*: No activity against $H_{37}Ra$ *Cytotoxic activity*: No activity against KB, NCI-H187, MCF7	[48]
A. odoratus	**384**	*Antitubercular activity*: No activity against $H_{37}Ra$ *Cytotoxic activity*: No activity against KB, NCI-H187, MCF7	[48]
Pseudobaeospora pyrifera	**385**	*Anticholinesterase activity*: 20% at 100 μM aginst AChE	[141]
P. pyrifera	**386**	*Anticholinesterase activity*: 7% at 100 μM aginst AChE	[141]
Macrolepiota neomastoidea	**392**	*Antimycobacterial activity*: MIC > 64.0 μg/mL against *Mycobacterium tuberculosis* $H_{37}Rr$ *Cytotoxic activity*: No activity against Vero cells	[59]
Hericium erinaceum	**406**	*Cytotoxic activity*: IC_{50}: 2.9 μM against HCT-15 IC_{50}: 1.9 μM against SK-MEL-2 IC_{50}: 3.1 μM against SK-OV-3 IC_{50}: 2.6 μM against A549	[146]

Continued

TABLE 5 The Biologically Active Alkaloids and Their Activity Values—Continued

Mushroom Species	Isolated Alkaloids	Bioactivities Tested with Activity Values	References
H. erinaceum	**407**	*Cytotoxic activity*: IC_{50}: 19 μM against HCT-15 IC_{50}: 3.1 μM against SK-MEL-2 IC_{50}: 8.9 μM against SK-OV-3 IC_{50}: 21 μM against A549	[146]
H. erinaceum	**408**	*Cytotoxic activity*: IC_{50}: 14 μM against HCT-15 IC_{50}: 7.7 μM against SK-MEL-2 IC_{50}: 11 μM against SK-OV-3 IC_{50}: 11 μM against A549	[146]

Ganoderma sinense Zhao, Xu et Zhang the alkaloid sinensine (**381**) has been isolated. Compound **381** has been reported to show a biological activity in protecting H_2O_2-oxidation induced injury on HUVEC (human umbilical vein endothelial cells). The EC_{50} value was found to be 6.23 mmol/L [138].

Nicotinic acid (**382**), hypaphorine (**383**), and 5-hydroxyhypaphorine (**384**) have been isolated from the *A. odoratus* by *Arpha et al*. [48] as well. The antibacterial activity against *M. tuberculosis* $H_{37}Ra$ and the cytotoxicity against MCF-7, KB and NCI-H187, cancerous cell lines of compounds **383**–**384** were evaluated and found that both were inactive in both assays [48].

Pseudobaeospora pyrifera which belongs to Tricholomataceae family is a rare mushroom species [139]. From the fruiting body of *P. pyrifera*, three alkaloids (**385**–**387**), named pyriferines A–C were isolated [140]. *Rhee et al*. [141] assayed these compounds against acetylcholinesterase which is a chief enzyme in Alzheimer's disease. At 100 μM concentration pyriferines A and B (**385**, **386**) inhibited enzyme 20% and 7%, respectively [141]. The poisonous mushroom species Macrolepiota neomastoidea Hongo (Agaricaceae) contains γ-lactam alkaloids, lepiotins A (**388**), B (**389**), C (**390**), and (*R*)-5-hydroxypyrrolidine-2-one (**391**) [122]. As determined by *Stanikunaite et al*. [59] using bioactivity-controlled fractionation, the ethyl acetate extract of *A. pteridis* which looks like a truffle species, was found to contain *N*,*N*,*N*-trimethylphenylalanine (**392**) which demonstrated no cytotoxicity to African

green monkey kidney fibroblast cells up to 100 μg/mL. The compound (**392**) was also tested for its tuberculosis activity against *M. tuberculosis*. Since the determined MIC value was greater than 64 μg/mL, it was considered as inactive [59].

The *Mycena haematopus* commonly known as the bleeding fairy helmet mushroom was studied by *Peters et al*. [142]. From this mushroom four red alkaloid pigments, haematopodin B (**393**) and the mycenarubins D (**395**), E (**397**) and F (**399**) were isolated. In addition to these isolated red pigments (**393**, **395**–**399**), haematopodin (**394**) were also isolated from the same mushroom. According to *Peters et al*., haematopodin (**394**) is the degradation product of haematopodin B (**393**), which is the native main pigment of *M. haematopus* [142].

A group from Garching city Germany studied three *Mycena* species and determined the presence of pyrroloquinoline type alkaloids (**395**–**398**). These alkaloids have been previously isolated from the marine products; yet, *Peters et al*. [142] and Peters and Spiteller [143,144] discussed that both marine products and *Mycena* mushroom species contain pyrroloquinoline type alkaloids [142–144]. However, there have been no biological activity studies to date on the isolated alkaloids (**395**–**398**).

Sanguinone A (**400**), decarboxydehydrosanguinone A (**401**), and sanguinone B (**402**), which are blue alkaloid pigments, along with sanguinolentaquinone (**403**), which are red indoloquinone alkaloid were isolated from the *Mycena sanguinolenta* (the bleeding bonnet) [143–144]. Mycenarubin A (**404**) and a dimer mycenarubin B (**405**) which are also called as red alkaloid pigments have been isolated from the mushroom *Mycena rosea*, (the rosy bonnet) [143–144]. The *o*-quinoid structured mycenarubin A (**404**) was inactive against *E. coli*, *B. brevis*, or *B. subtilis*, and *Cladosporium cucumerinum* [143]. However, the *para*-iminoquinone-structured pyrroloquinoline alkaloids are responsible for the various bioactivities [145]. Mostly, the pyrroloquinoline alkaloids such as the batzellines, damirones, discorhabdins, isobatzellines, makaluvamines, and wakayin are isolated from the marine sources. However, sanguinones whose structure resemble to discorhabdins were isolated from *M. sanguinolenta* mushroom species [144].

H. erinaceum was considered as a potential anticancer mushroom species since its extracts showed antitumor activity against HCT-15, SK-MEL-2, SK-OV-3, and A549 cancerous cell lines. The bioactivity-guided fractionation of *H. erinaceum* afforded three alkaloids; namely, isohericenone (**406**), isohericerin (**407**), and erinacerin A (**408**). The cytoxicity of isolated alkoaloids to HCT-15, SK-MEL-2, SK-OV-3, and A549 were also studied. Isohericenone (**406**) indicated better IC_{50} values than that of isohericerin (**407**) against all cell lines studied. Alkaloids **406** and **407** indicated cytotoxic activity against HCT-15 (IC_{50}: 2.9 and 19 μM, respectively), SK-MEL-2 (IC_{50}: 1.9 and 3.1 μM, respectively), SK-OV-3 (IC_{50}: 3.1 and 8.9 μM, respectively), and A549 (IC_{50}: 2.6 and 21 μM, respectively)

cancerous cell lines. Erinacerin A (**408**) showed anticancer activity against the same cancerous cell lines. The IC_{50} values were between 7.7 and 14 μM. At the same conditions against same cancerous cell lines the doxorubicin indicated 0.081, 0.002, 0.003, and 0.001 μM IC_{50} values, respectively [146].

Another class of alkaloids isolated from the mushroom *Boletus curtisii* are the canthin-6-one derivatives. These compounds are novel sulfurous alkaloid compounds [147]. This class of alkaloids has been previously isolated from the higher plants and *Brockelmann et al*. [147] proved that these compounds can be found in this mushroom species. The structures of the compounds were determined as: 2-Methyl-*β*-carbolinium-1-propanoate (**409**), *β*-carboline-1-propanoic acid (**410**), 9-deoxycurtisin (**411**), curtisin (**412**), 11-(methylthio)canthin-6-one (**413**), 9-(methylthio)canthin-6-one (**414**), canthin-6-one (**415**), 4-(methylthio)canthin-6-one (**416**), and 5-(methylthio) canthin-6-one (**417**) [147].

G. lucidum has been the most studied *Ganoderma* species due to its anticancer and other biological activities. It is a rich source of various compounds including polysaccharides, terpenes, and steroids. The pyrrole type of alkaloids ganodine (**418**), ganoine (**419**) as well as N-(2'-methylpentyl)-5-hydroxymethyl-pyrrylaldehyde (**420**) have also been isolated from the *G. lucidum* [148].

Other Miscellaneous Bioactive Compounds

The methanol extract of the basidiomycete of *S. joedes* has been shown to contain a cerebroside (cerebroside B (**421**)) [107]. The cytotoxic activity of this compound was also evaluated by the same group. However, no activity indicated [107]. Cerebroside B (**421**) is a sphingolipid and sphingolipids have been reported to regulate arrest of cell growth, differentiation, apoptosis, and senescence [105,107].

Another cerebroside (**422**) was also isolated from the mushroom *S. scabrosus* [77]. The isolated compound (**422**) was tested for its effect of NGF-induced neurite outgrowth on PC12 rat pheochromocytoma cell line, however, no activity was found [77].

The dichloromethane extract of an edible mushroom *H. marmoreus* also yielded eight polyisoprenepolyols (**423–430**). Their antitubercular activity against *M. tuberculosis* strain $H_{37}Rv$ was investigated [106]. Two polyisoprenepolyols (**423**, **427**) showed activity, while the others were inactive. In addition, three polyisoprenepolyols (**423**, **425–426**) were also investigated antiviral activity against antigen activation induced Epstein–Barr virus. In Raji cells, this virus is known as the tumor promoter. However, no more activity was found [106].

Lectins are another class of bioactive compounds. They are available in mushrooms together with animals, plants, bacteria, viruses, and other

fungi. Recently, lectins have been isolated from many mushroom species such as *Schizophyllum commune* [149], *V. volvacea* [150,151], *G. lucidum* [152], *F. velutipes* [153], *G. frondosa* [154], *Tricholoma mongolicum* [155], and *Agaricus bisporus* [156], *T. mongolicum* [157,158], *Agrocybe cylindracea* [159,7]. The isolated lectins from various biological sources were tested for their different bioactivities. The bioactivity data revealed that they exhibited anticancer, antitumor/cytotoxic [155,159], antiproliferative [156], immunomodulatory [155], mitogenic [160], and hypotensive [157] activities.

CONCLUSION

In the last two decades, Mushroom species including *Ganoderma* genus have been investigated to obtain bioactive compounds. Most of the compounds isolated were lanostanoids or polysaccharides. Mostly, the cytotoxicity of lanostanoids was tested against various cancerous cell lines as given in Table 2. The anticancer, antimalarial, anticholinesterase, antiviral, antibacterial, and anti-inflammatory activities were also tested for these compound classes. The polysaccharides, particularly β-glucans known to strengthen the immune system were also isolated and tested for their different bioactivities (Table 1). As for the mushroom steroids, they exhibited anticomplement, antimicrobial, cytotoxic activity, antitubercular, and antiviral activities (Table 3). The phenolic compounds, however, are important for human health, particularly to treat cardiovascular, cancer, diabetes, Alzheimer's, and Parkinson's diseases.

Even though than the compounds isolated from various mushroom species indicating strong activities, it does not mean that all mushrooms are always nostrums [161]. Particularly, the medicinal uses of *Ganoderma* species' are need to be optimized for their production. Thus, their ingredients can become stabile in future. Then they may be used to treat some of the diseases including cancer without any doubt. Because the chemical compounds of the species may be affected by the altitude, humidity, growth place, temperature, and etc. On the other hand, the synergic effects of the compounds can be responsible for the disease healing.

Nevertheless, in order to discover new knowledge continuing studies with the mushrooms including *Ganoderma* species are even necessary, in the future. As seen from the whole review, most of the activities tested on the isolated compounds are *in vitro* studies. This may be resulted due to the limited amount of natural products isolated. Hence, the *in vivo* studies should also be carried out to understand the mechanisms.

Overall, this chapter focused on providing a comprehensive summary of the compounds that have been isolated from many different mushroom species which include polysaccharides, terpenoids, steroids, phenolics, and alkaloids and provided evidence of their bioactivities (Fig. 1).

THE POLYSACCARIDES

→ 3)-β-D-Glc*p*-(1→ 3)-β-D-Glc*p*-(1→ 6)-β-D-Glc*p*-(1→ 6)-β-D-Glc*p*-(1→
6
↑
1
β-D-Glc*p*

1

Z **Y** **Z**
→ 3)-β-D-Glc*p*-(1→ 3)-β-D-Glc*p*-(1→ 3) -β-D-Glc*p*-(1→
6
↑
1
β-D-Glc*p*
X

→ 6)-β-D-Glc*p*-(1→

β-glucan PS-I
Water soluable glucan

3

β-glucan PS-II
Water insoluable glucan

4

B **C** **D** **D**
→ 3)-β-D-Glc*p*-(1→ 3)-β-D-Glc*p*-(1→ 3)-β-D-Glc*p*-(1→ 3)-β-D-Glc*p*-(1→
6
↑
1
β-D-Glc*p*
A

5

E **D** **E** **C**
→ 3)-β-D-Glc*p*-(1→ 3)-β-D-Glc*p*-(1→ 3)-β-D-Glc*p*-(1→ 3)-α-D-Glc*p*-(1→
4
↑
1
α-D-Glc*p*
B

→ 6)-β-D-Glc*p*-(1→

β-glucan PS-1

6

β-glucan PS-2

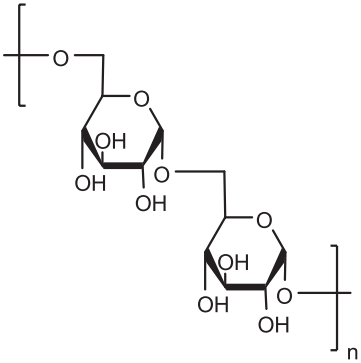
7

C E C D G B H B
→ 3)-α-D-Man*p*-(1→ 3)-β-D-Gal*p*-(1→ 3) -α-D-Man*p*-(1→ 3)-α-D-Gal*p*-(1→ 6) -α-D-Gal*p*-(1→ 2)-α-L-Fuc*p*-(1→ 6) -α-D-Glc*p*-(1→ 2)-α-L-Fuc*p*-(1→
2 2
↑ ↑
1 1
α-D-Glc*p* β-D-Glc*p*
A F

9

→3)-α-D-Glcp-(1→3)-β-D-Glcp-(1→3)-β-D-Glcp-(1→6)-β-D-Glcp-(1→
6
↑
1
α-D-Glcp

10



11

→ 3)-β-D-Glcp-(1→ 3)-β-D-Glcp-(1→ 3)-β-D-Glcp-(1→ 6)-β-D-Glcp-(1→
6
↑
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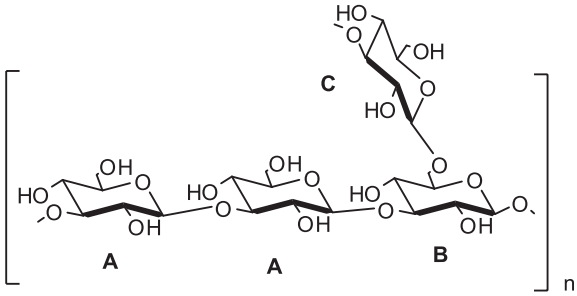
β-D-Glcp

12



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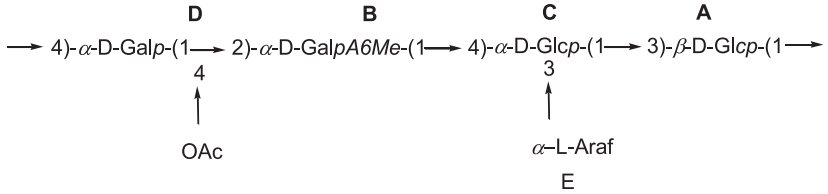
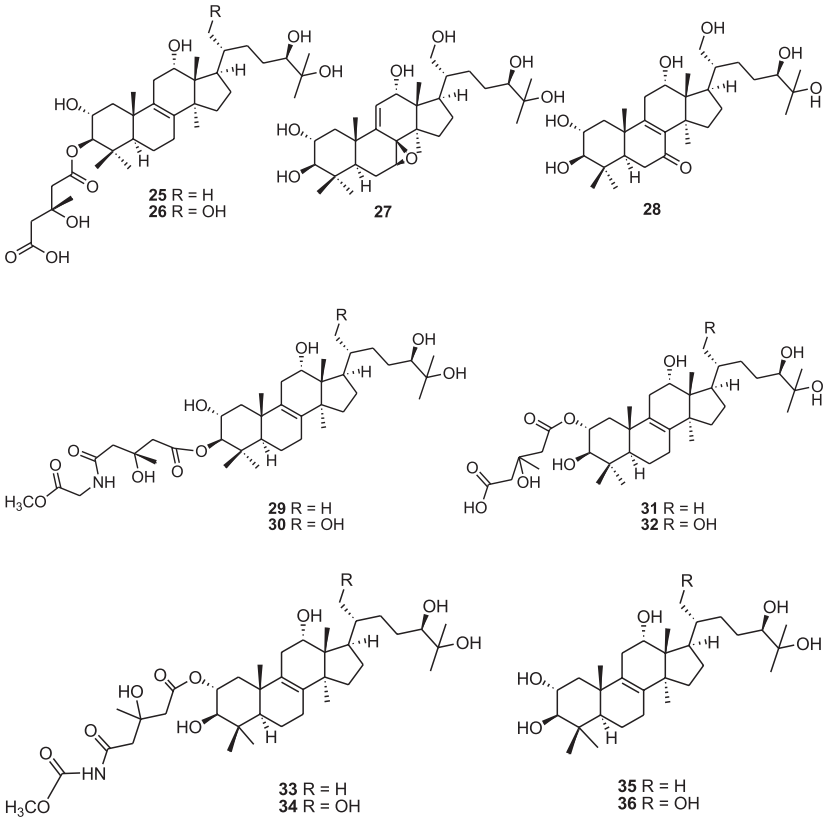
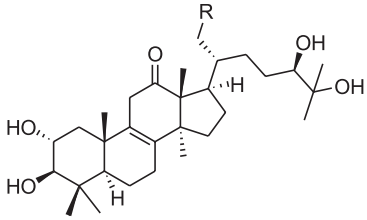


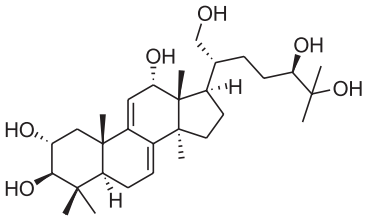
16

THE TERPENOIDS





37 R = H
38 R = OH



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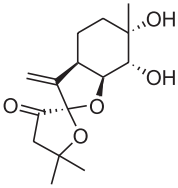

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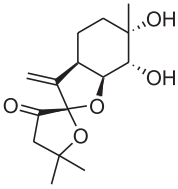

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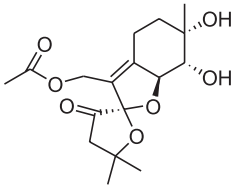

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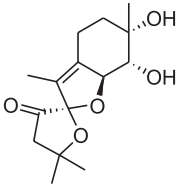

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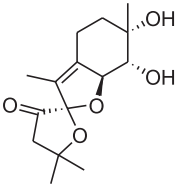

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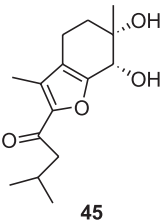

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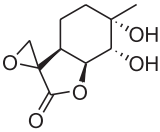

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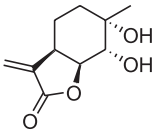

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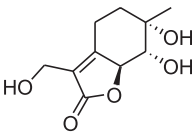

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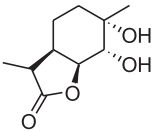

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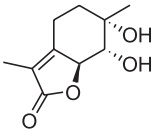

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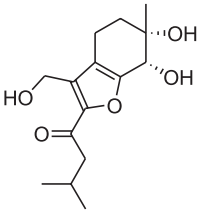

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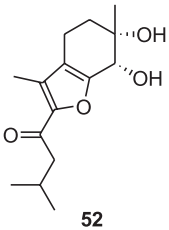

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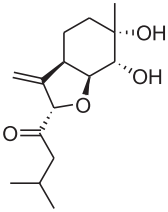

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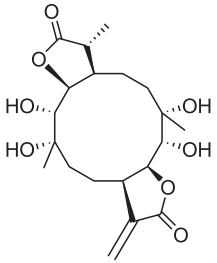

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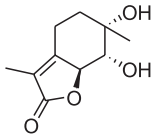

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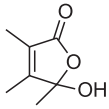

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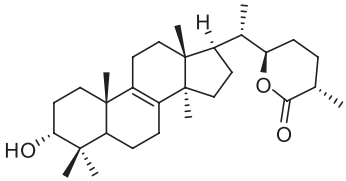

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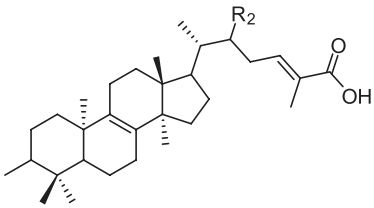
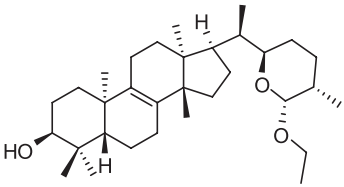


58 R_1 = O, R_2 = α-OAc
59 R_1 = O, R_2 = α-OH
60 R_1 = α-OH, R_2 = α-OH
61 R_1 = α-OH, R_2 = α-OAc



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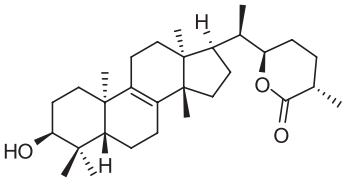

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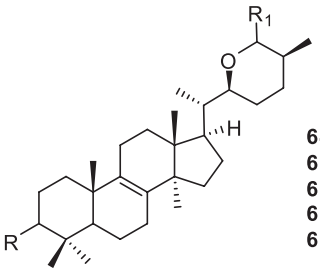
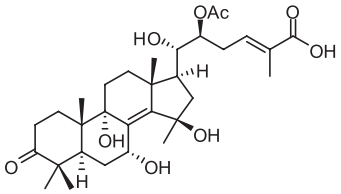


64 R= α-OH R_1= O
65 R= R_1= O
66 R=O R_1= α-OH
67 R= β-OH R_1= α-OH
68 R= R_1= α-OH



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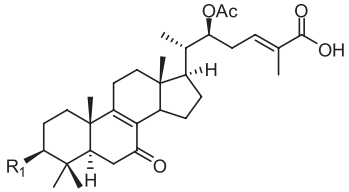
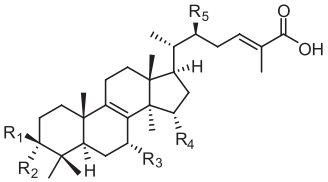
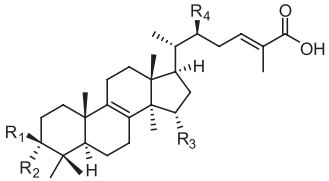


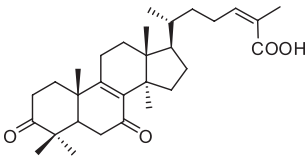
70 R_1= OAc
71 R_1= OH



72 R_1, R_2 = O, R_3 = OH, R_4 = OAc, R_5 = OAc
73 R_1, R_2 = O, R_3 = OH, R_4 = H, R_5 = OAc
74 R_1, R_2 = O, R_3 = OCH_3, R_4 = H, R_5 = OAc
76 R_1, R_2 = O, R_3 = OH, R_4 = OAc, R_5 = H
77 R_1 = H, R_2 = OAc, R_3 = OCH_3, R_4 = OAc, R_5 = OAc



75 R_1, R_2 = O, R_3 = H, R_4 = OAc
78 R_1, R_2 = O, R_3 = OAc, R_4 = OAc
79 R_1, R_2 = O, R_3 = OAc, R_4 = H
80 R_1, R_2 = O, R_3 = OH, R_4 = OAc
81 R_1 = OAc, R_2 = H, R_3 = OAc, R_4 = OAc
82 R_1 = OH, R_2 = H, R_3 = OAc, R_4 = OAc
83 R_1 = H, R_2 = OAc, R_3 = OAc, R_4 = OAc
84 R_1 = H, R_2 = OAc, R_3 = H, R_4 = OAc
85 R_1 = H, R_2 = OH, R_3 = OAc, R_4 = OAc
86 R_1 = H, R_2 = OH, R_3 = OAc, R_4 = H
87 R_1 = H, R_2 = OH, R_3 = H, R_4 = OAc



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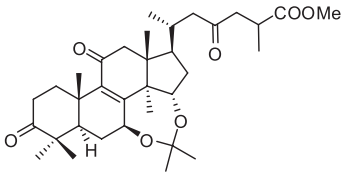

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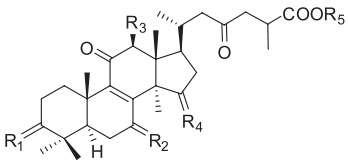
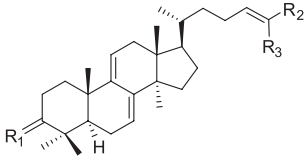
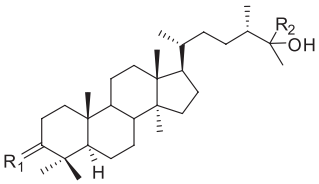


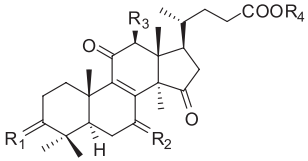
	R_1	R_2	R_3	R_4	R_5
90	αH, βOH	O	OAc	O	Bu
91	O	αH, βOH	H	αH, βOH	Me
92	αH, βOH	αH, βOH	H	O	H
93	O	O	H	O	H
94	O	H_2	H	αH, βOH	H



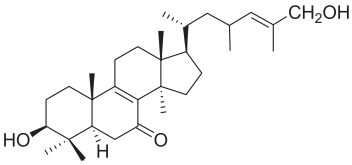
	R_1	R_2	R_3
95	αH, βOH	O	OAc
96	O	αH, βOH	H
97	αH, βOH	αH, βOH	H



	R_1	R_2
98	αH, βOH	Me
99	O	Me
100	O	CH_2OH



	R_1	R_2	R_3	R_4
102	αH, βOH	αH, βOH	H	H
103	O	αH, βOH	H	Me
104	αH, βOH	O	OAc	Me
105	αH, βOH	αH, βOH	H	H
106	O	αH, βOH	H	Me



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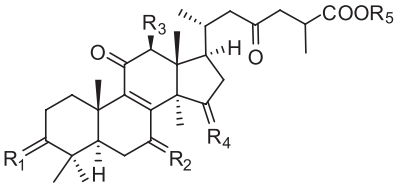
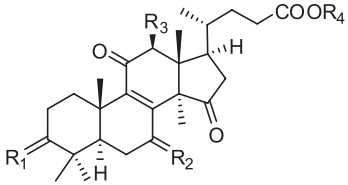
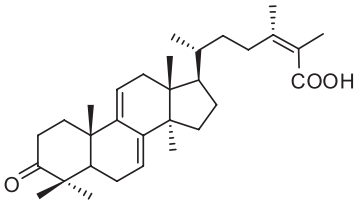




	R_1	R_2	R_3	R_4	R_5
107	O	βOH	H	αOH	Bu
108	βOH	βOH	H	O	Bu
109	O	βOH	H	αOH	H
110	βOH	βOH	H	O	Me
111	O	βOH	H	O	Me
112	O	O	H	O	H
113	βOH	O	OAc	O	Me

	R_1	R_2	R_3	R_4
114	O	βOH	H	Me
115	βOH	βOH	OAc	Me
116	O	O	H	Me



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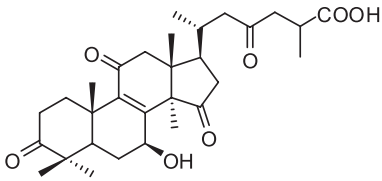

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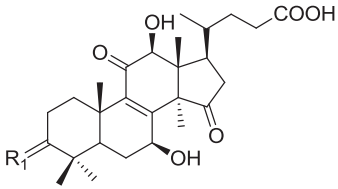
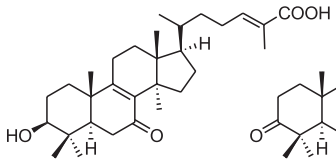


119 R_1= O
120 R_1= β-OH



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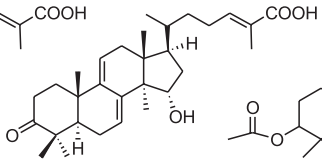

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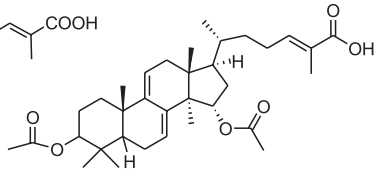

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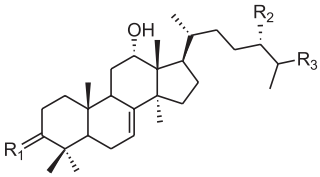
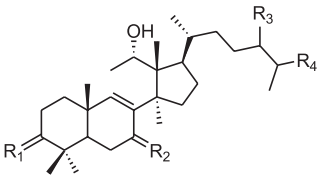
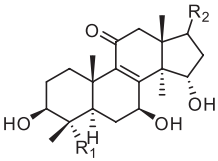
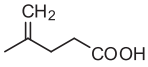
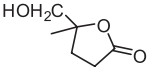


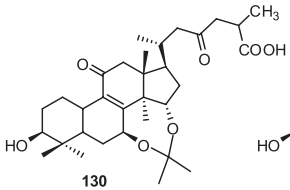


	R_1	R_2	R_3		R_1	R_2	R_6	R_8
124	OH, H	$\Delta^{24,25}$	CHO	**125**	O	O	$\Delta^{24,25}$	CHO
127	O	$\Delta^{24,25}$	CH_2OH	**126**	OH, H	O	$\Delta^{24,25}$	CHO



128 R_1= CH_2OH, R_2= 

129 R_1= CH_3 R_2= 



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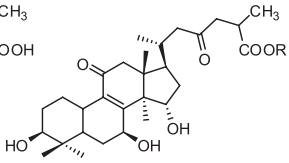
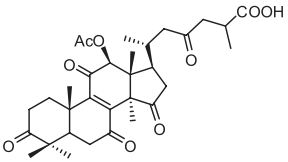


131 R=CH_3
132 R=H



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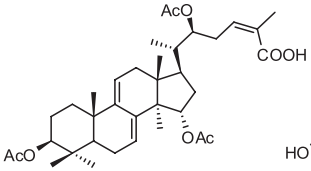

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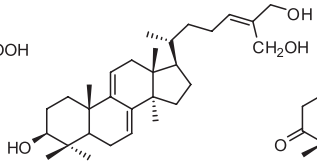

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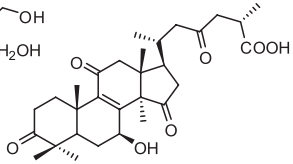

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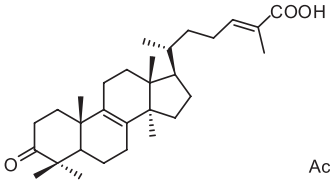

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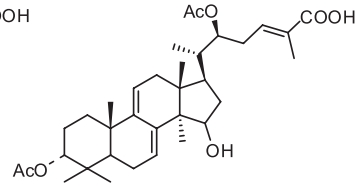

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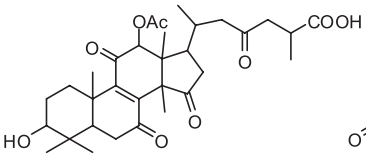

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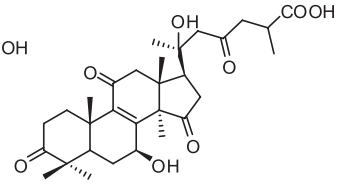

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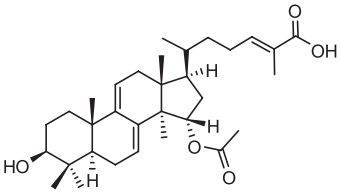

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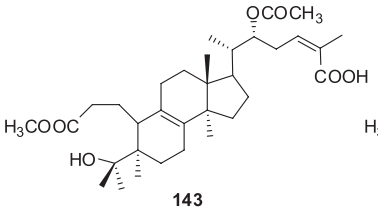

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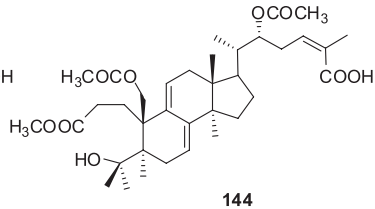

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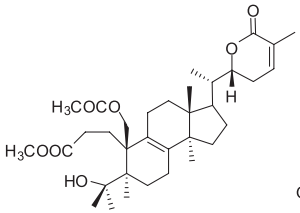

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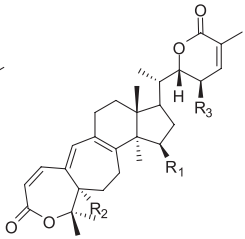
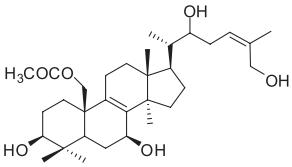
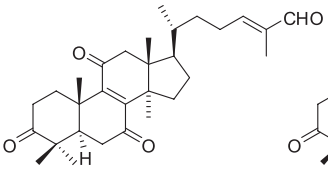




150

146 R_1= OAc R_2= H R_3=OH
147 R_1= OAc R_2= H R_3= H
148 R_1= H R_2= H R_3=H
149 R_1= OAc R_2= OH R_3= H



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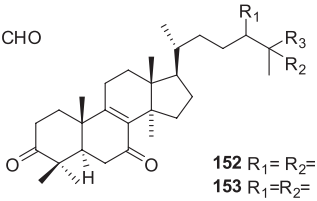
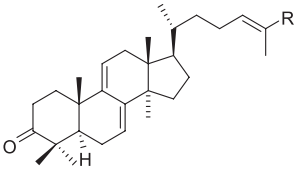
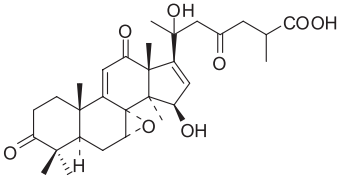


152 R_1= R_2= $\Delta^{24(25)}$, R_3= CH_2OH
153 R_1=R_2= -O- , R_3= CH_2OH
154 R_1=R_2= $\Delta^{24(25)}$, R_3= CHO



155 R= CH_2OH
156 R= CHO



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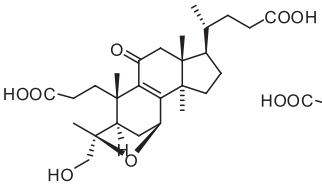

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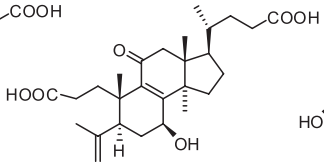

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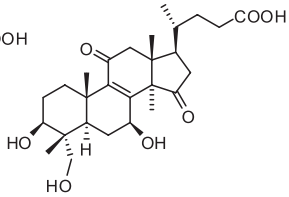

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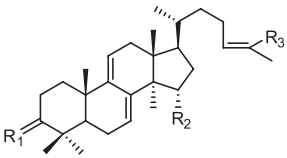
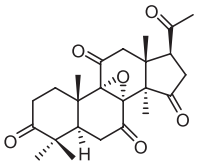


161 R_1= β–OH, H; R_2= H; R_3= CHO
162 R_1= O; R_2= α–OH; R_3= CH_2OH
164 R_1= O; R_2= H; R_3= CHO



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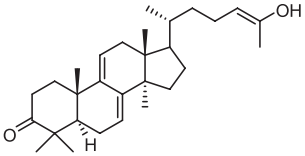

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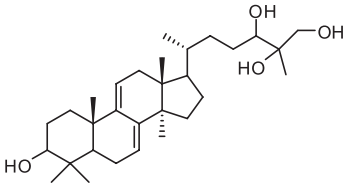

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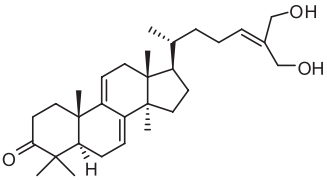

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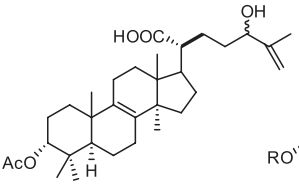

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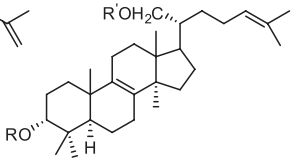
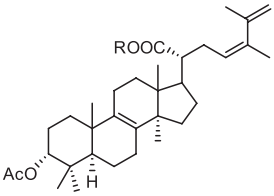
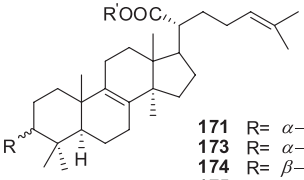


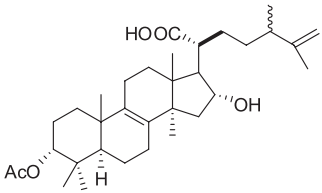
169 R= Ac, R'= β–D-xylosyl



170 R= β–D-xylosyl



171 R= α–OAc, R'= H
173 R= α–OAc, R'= β–D-glucosyl
174 R= β–OH, R'= H
175 R= O, R'= H



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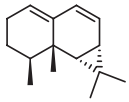

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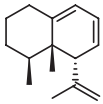

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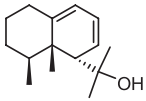

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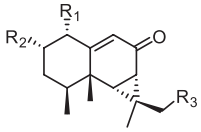
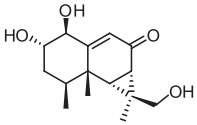


179 R_1 = H; R_2 = H; R_3 = H
180 R_1 = H; R_2 = OH; R_3 = H
181 R_1 = OH; R_2 = OH; R_3 = H
182 R_1 = H; R_2 = OH; R_3 = OH
183 R_1 = OH; R_2 = OH; R_3 = OH



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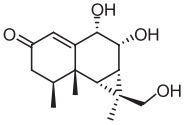

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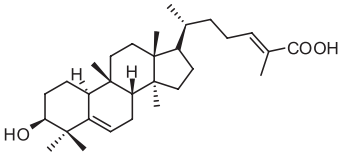

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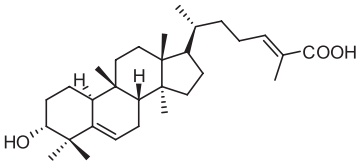

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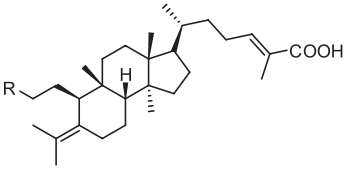
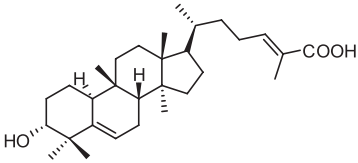


188 R = COOH
189 R = CH_2OH



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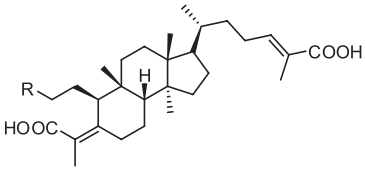
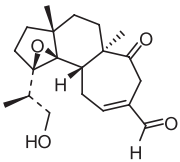


188 R = COOH
189 R = CH_2OH



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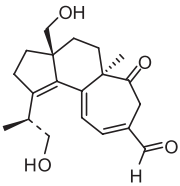

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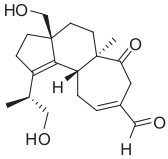

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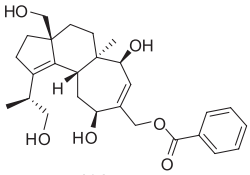

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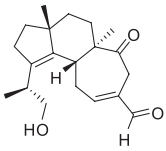

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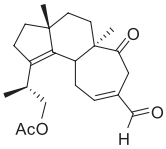

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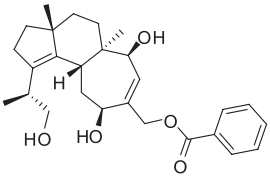

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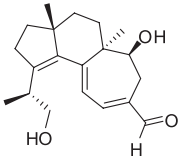

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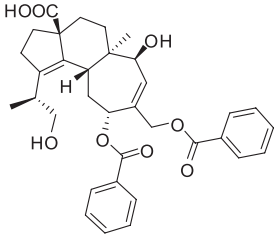

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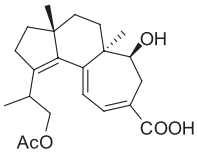

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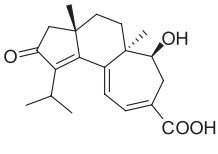

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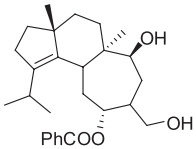

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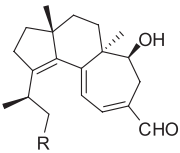
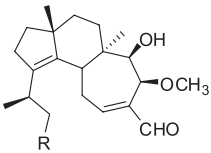
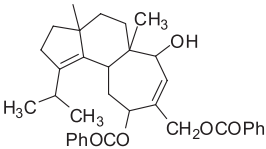


203: R=OH
205: R=O-linoleoyl
206: R=O-oleoyl
207: R=O-stearoyl
208: R=H
210: R=$OCOC_7H_{15}$
211: R=$OCOC_3H_7$
212: R=$OCOCH_3$
213: R=$OCOC_6H_5$
214: R=$OCOC(CH_3)_3$



204: R=H
209: R=OH



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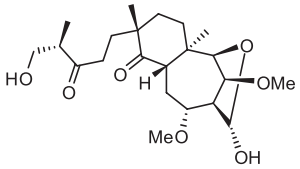

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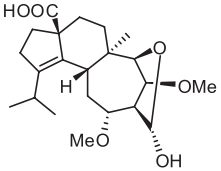

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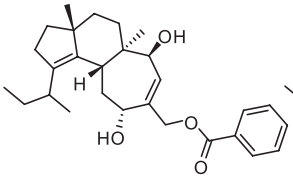

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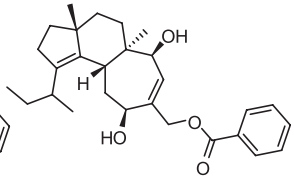

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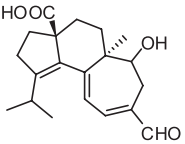

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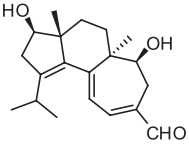

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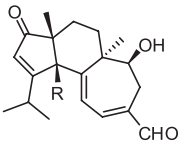
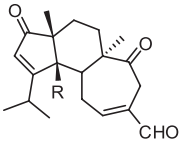
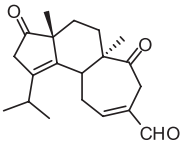


222 R= OH
223 R= H



224 R= H
225 R= OH



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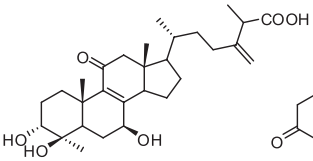

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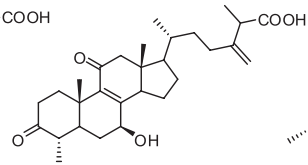

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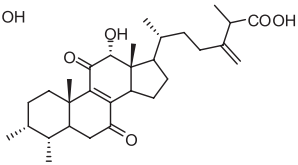

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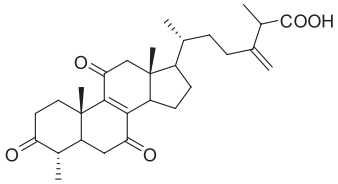

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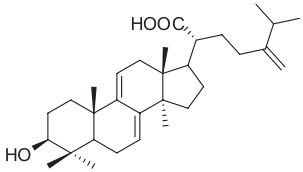

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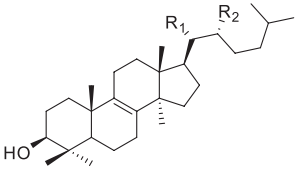

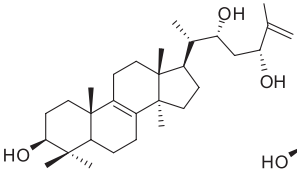

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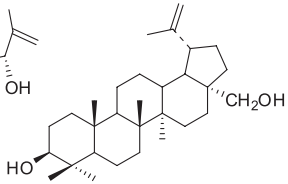
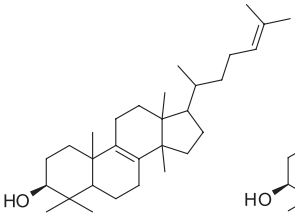


237

232: $R_1 = CH_3$, $R_2 = H$
233: $R_1 = CHO$, $R_2 = H$
234: $R_1 = CH_3$, $R_2 = OH$
235: $R_1 = COOH$, $R_2 = H$



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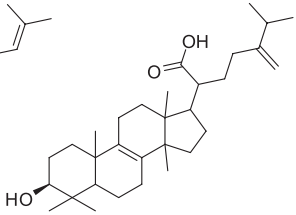

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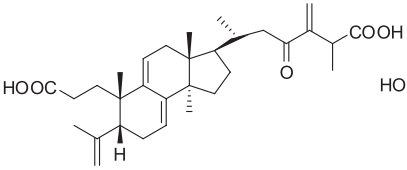

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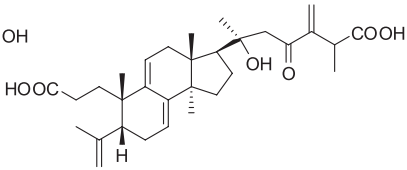

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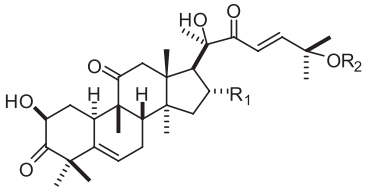
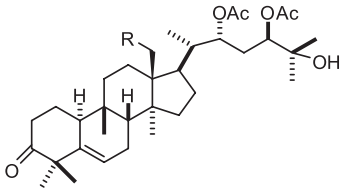
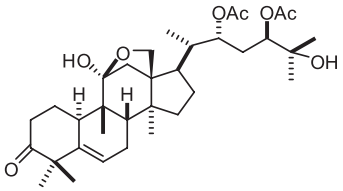


242 R_1= OH; R_2= Ac
243 R_1= Oleoyloxy, linoleoyloxy, palmitoyloxy; R_2= Ac
246 R_1= OH; R_2= H
247 R_1=H; R_2= Ac



244 R= OH
248 R= H



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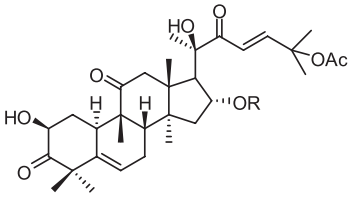
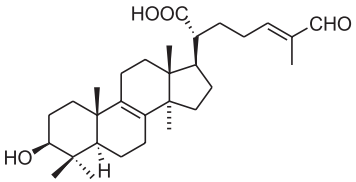


249 R= H
250a R= Oleyl
250b R= Linoleyl
250c R= Palmityl



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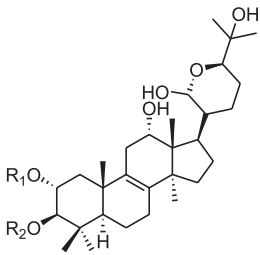
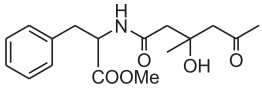
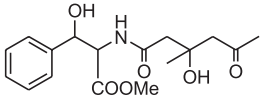


252 R_1= H, R_2 =

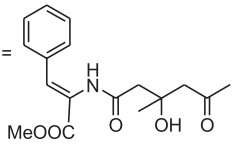


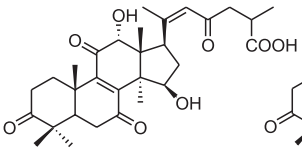
253 R_1 =



R_2 = H

254 R_1 = Ac, R_2 =





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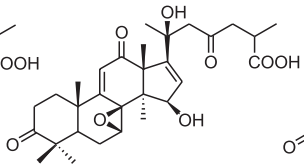

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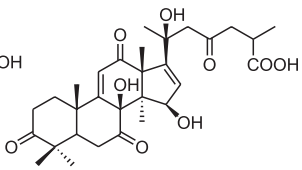

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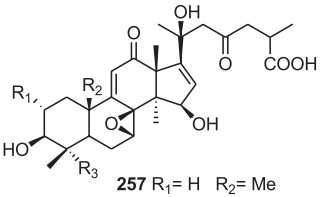
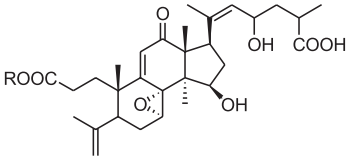
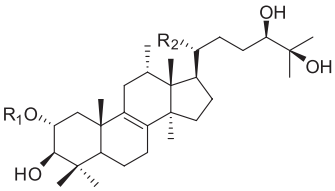


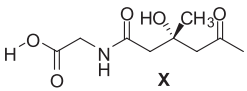
257 R_1= H R_2= Me R_3= Me
258 R_1= OH R_2= Me R_3= Me
259 R_1= H R_2= Me R_3= CH2OH
260 R_1= H R_2= CH2OH R_3= Me



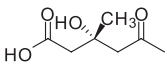
262 R= H
263 R= Me



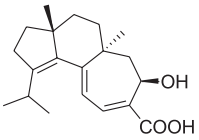
264 R_1= X; R_2=CH_2OH
265 R_1= X; R_2= CH_3
266 R_1= Y; R_2= CH_2OH



X



Y



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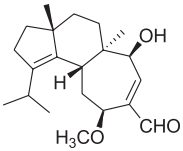

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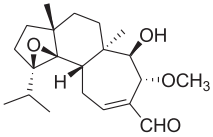

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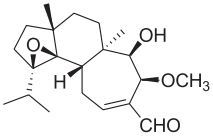

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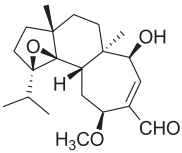

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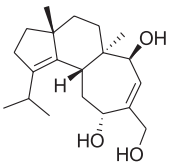

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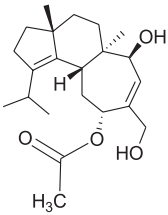

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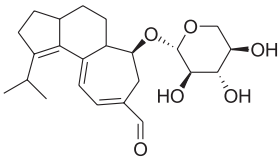

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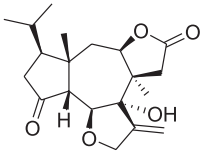

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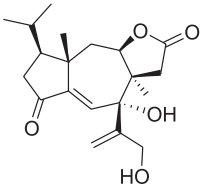

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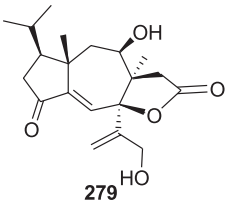

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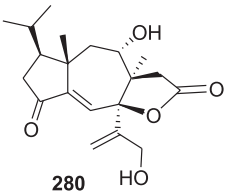

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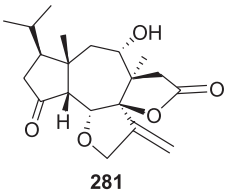

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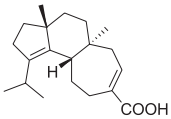

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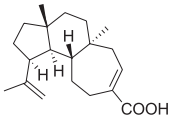

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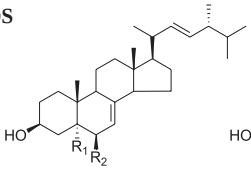
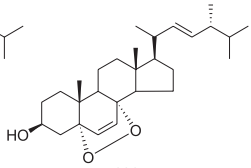
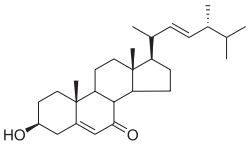
THE STEROIDS





286

284 R_1 =OH, R_2 =OMe
285 R_1 =R_2 =H



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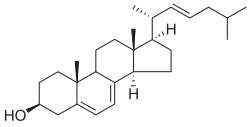

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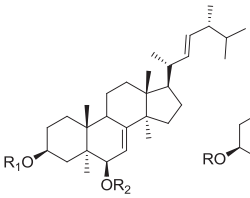
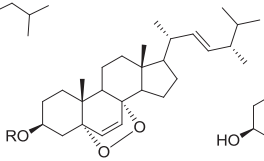
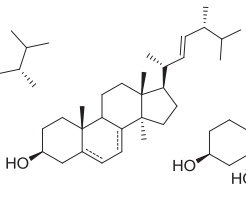


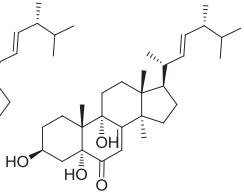
289 R_1=Glc, R_2=H
292 R1=H, R_2=Me



290 R=Glc



291 Δ^7



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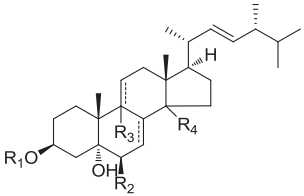
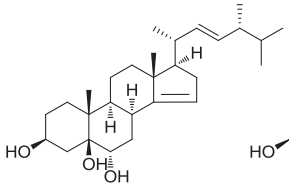


	R_1	R_2	R_3	R_4	
294	H	β–OH	-	α–OH	$\Delta^{7,9(11)}$
296	H	β–OEt	αH	αH	Δ^7
299	H	α–OH	αH	αH	Δ^7
300	H	β–OH	-	αH	$\Delta^{7,9(11)}$
301	H	β–OH	α–OH	αH	Δ^7
302	Glu	β–OH	αH	αH	Δ^7



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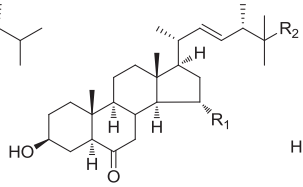
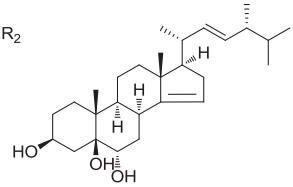


	R_1	R_2
297	O-Glu	OH
298	H	H



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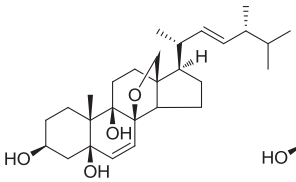

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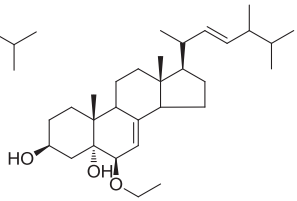

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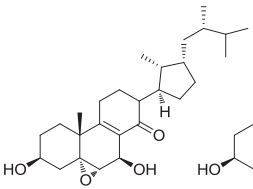

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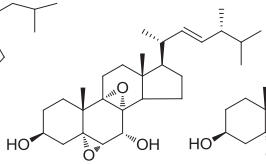

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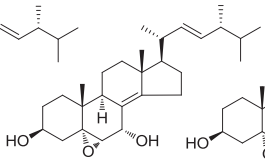

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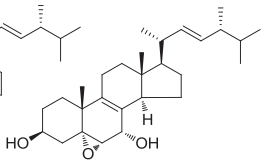

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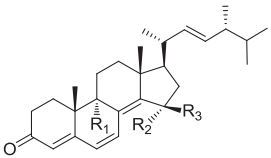
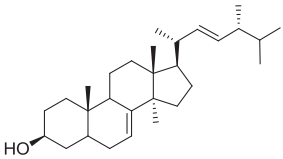


310 R_1 =H, R_2 =OH, R_3 =H
311 R_1 =H, R_2 =H, R_3 =OH
312 R_1 =OH, R_2, R_3 =O
313 R_1 =OH, R_2 =H, R_3 =H



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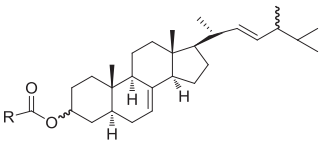
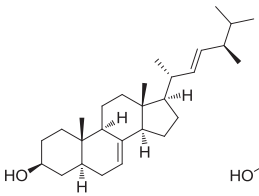


315 R= $-(CH_2)_{14}-CH_3$
316 R= $-(CH_2)_7-CH=CH-CH_2-CH=CH-(CH_2)_4-CH_3$
317 R= $-(CH_2)_{13}-CH_3$



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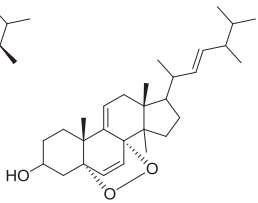

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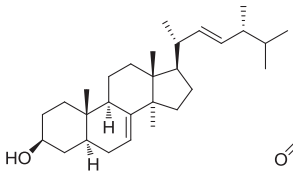

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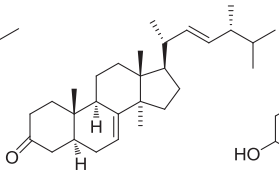

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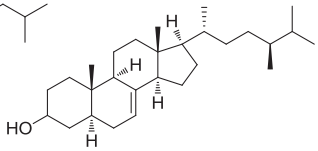

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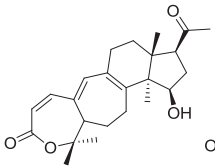

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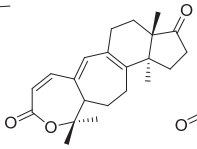

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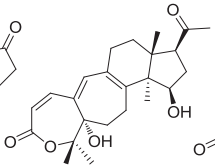

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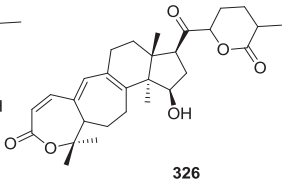

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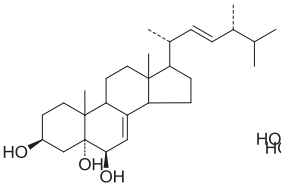

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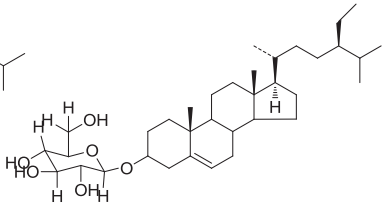

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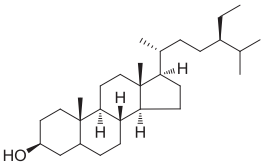

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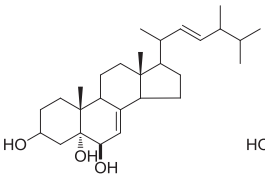

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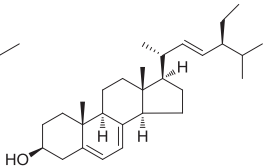

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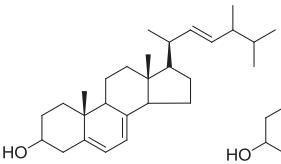

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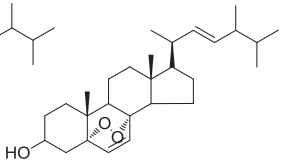

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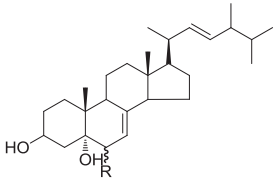
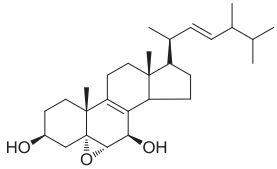
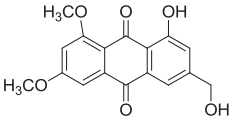


334 R= β–OH
335 R= α–OH
336 R= β–OH; 22,23 dihydro
337 R= β–OMe



338

THE PHENOLICS



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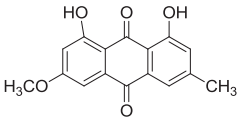

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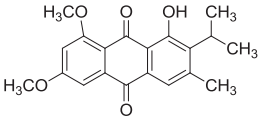

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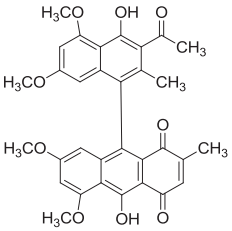

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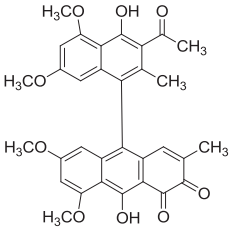

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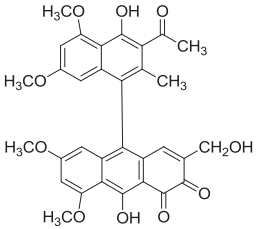

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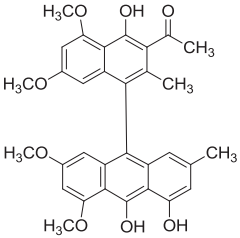

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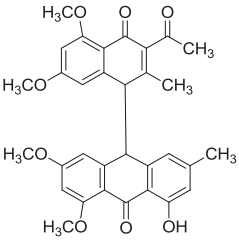

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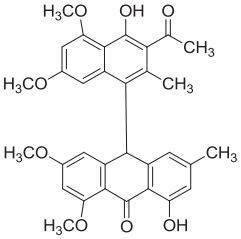

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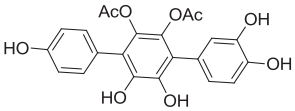

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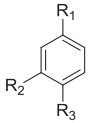
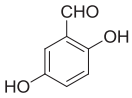


	R_1	R_2	R_3
349	CO_2H	H	H
350	CO_2Me	H	OH
351	CO_2H	OH	OH



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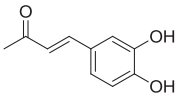

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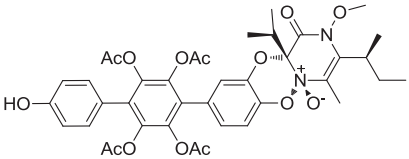

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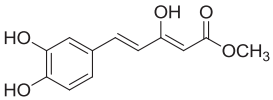

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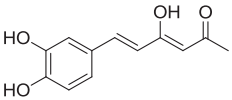

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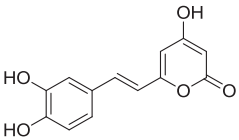

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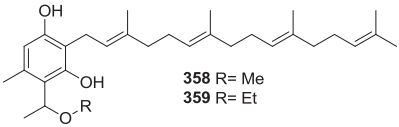
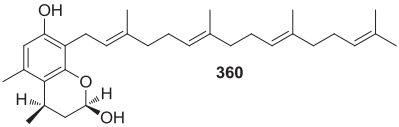


358 R= Me
359 R= Et



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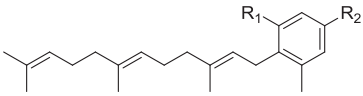
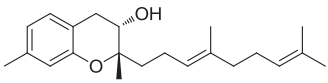


361 R_1= OH, R_2= Me
362 R_1= Me, R_2= OH



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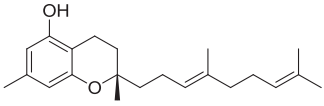

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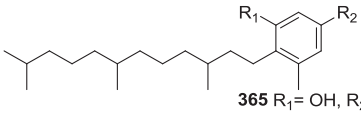
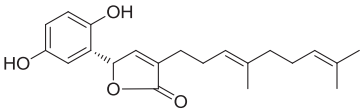


365 R_1= OH, R_2= Me
366 R_1= Me, R_2= OH



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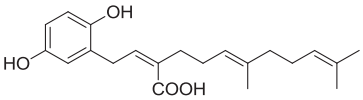

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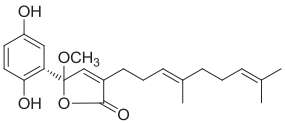

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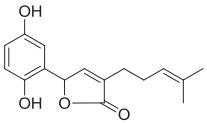

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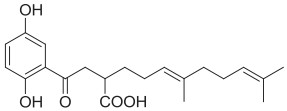

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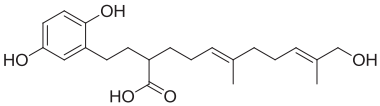

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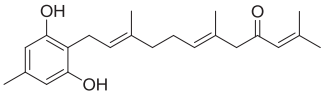

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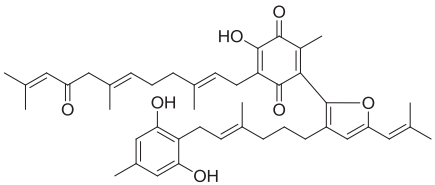

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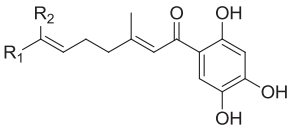
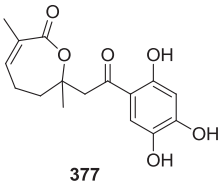


375 R_1 = CH_3, R_2 = COOH
376 R_1 = COOH, R_2 = CH_3



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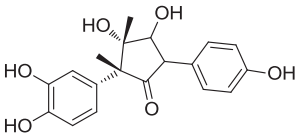

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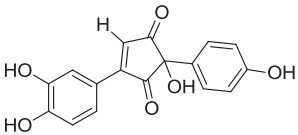
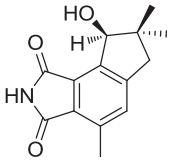


379

THE ALKALOIDS



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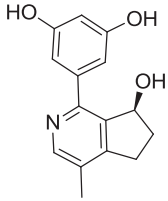

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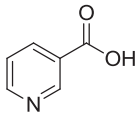

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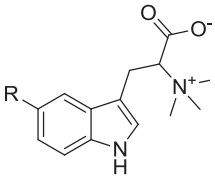
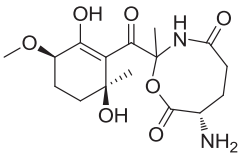


383 Hypaphorine
384 R= OH



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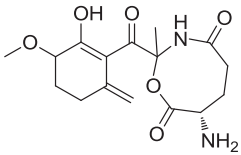

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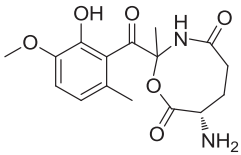

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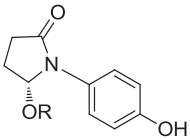
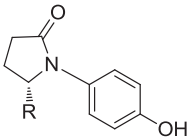
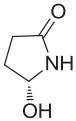


388 R = H
388r R = *(R)*-MTPA
388s R = *(S)*-MPTA



389 R = OCH_3
390 R = H_2



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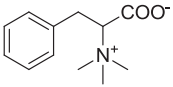

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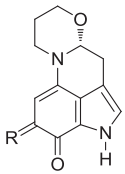
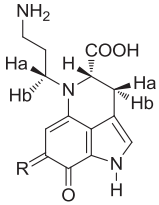
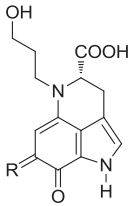


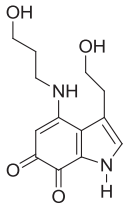
393 R= NH
394 R= O



395 R= NH
396 R= O



397 R= NH
398 R= O



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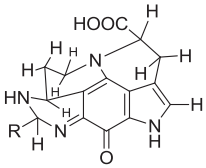
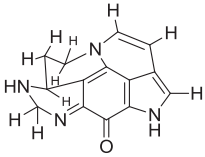


400 R = H
402 R = CH_3



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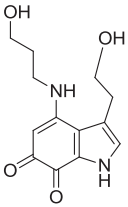

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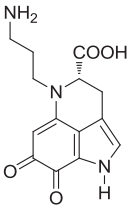

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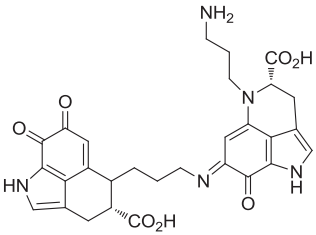

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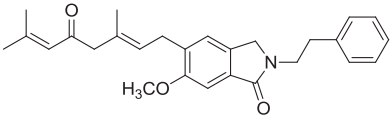

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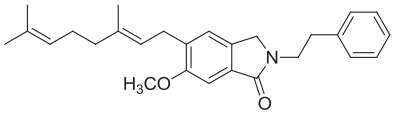

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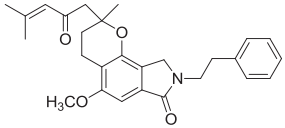

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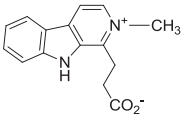

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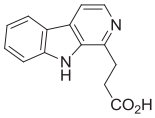

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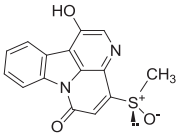

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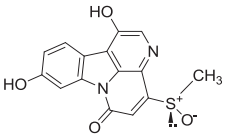

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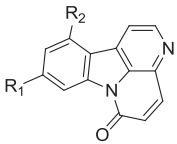
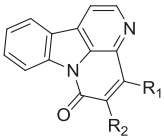
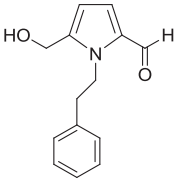


413 R_1= H, R_2= SMe
414 R_1= SMe, R_2= H



415 R_1= R_2= H
416 R_1= SMe, R_2= H
417 R_1= H, R_2= SMe



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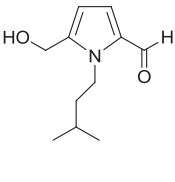

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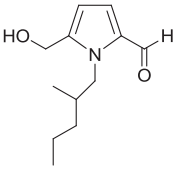

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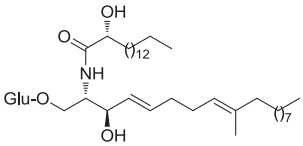
OTHERS



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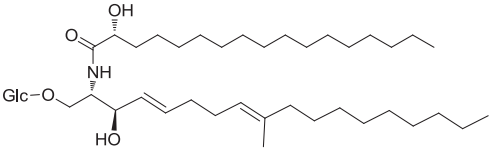

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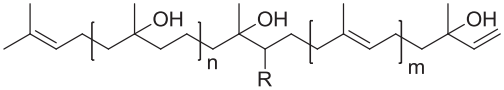

423 m= 1, n= 5, R=H
424 m= 2, n= 4, R=H
425 m= 1, n= 5, R=OH
426 m= 2, n= 5, R=H
427 m= 2, n= 5, R=OH
428 m= 1, n= 7, R=H
429 m= 1, n= 8, R=H
430 m= 1, n= 8, R=OH

FIGURE 1 The structures of compounds isolated from mushroom species.

LIST OF ABBREVIATIONS

1321N1	Human astrocytoma cells
A549	Nonsmall cell lung adenocarcinoma
AA	Arachidonic acid
ADP	Adenosine diphosphate
B16F1	Mouse melanoma cell lines
Bax	Proapoptotic Bcl-2 protein
Bel-7402	Human hepatocellular carcinoma cells
CAKI 1	Kidney carcinoma
CDK2	Cyclin-dependent kinase 2
CDK6	Cyclin-dependent kinase 6
ERK	Extracellular signal-regulated kinase
HCT116	Colon cancer cells
HCT-15	Colon adenocarcinoma
HEC-1	Endometrial adenocarcinoma cell line
HeLa	Human epitheloid cervix carcinoma
HeLa-S3	Cervix carcinoma
HepG2	Hepatocellular carcinoma
HIV	Human Immunodeficiency Virus
HL60	Human promyelocyeticleukemia cells
HMG-CoA	3-hydroxy-3-methylglutaryl-coenzyme A
HMV-1	Melanoma cell line
HOC-21	Ovarian serous cystadenocarcinoma cell line
HSV	Herpes simplex type I
HuH-7	Human hepatoma cell line
HUVEC	human umbilical vein endothelial cells
IARC	International Agency for Research on Cancer
IC_{50}	The half maximal inhibitory concentration
JNK	c-Jun *N*-terminal/stress-activated protein kinase
K562	Human erythromyeloblastoid leukemia cell line
KATO III	Stomach cancer cell
KB	Human epidermoid carcinoma cell
LC-MS	Liquid chromatography–mass spectrometry
LC-NMR	Liquid chromatography/nuclear magnetic resonance
LLC	Lewis lung carcinoma cell line
LO2	Steatotic hepatocytes cell line
LU99	Human lung cancer cell
MCF-7	Human breast cancer cell
MDP	Muramyl dipeptide protein
Meth-A	Murine sarcoma
MIC	Minimum Inhibitory Concentration
Mir-378	Mitogen-activated protein kinase pathway factor
MM-1CB	Melanoma cell line
MMP	Matrix metallo proteinase
MRP	Multidrug resistance-associated protein
MTT	3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide
NCI-H187	Human small cell lung cancer cell
NCI-H460	Human nonsmall cell lung cancer cells

NF-κB	The Nuclear Factor kappa B
NGF	Nerve growth Factor
NO	Nitric oxide
NUGC-3	Gastric carcinoma cell line
p53	Cellular tumor antigen
PAF-induced	Platelet-activating factor
PC12	Rat pheochromocytoma cell line
PMA	Phorbol-12-myristate-13-acetate
pRb	Retinoblastoma protein
RAW 264.7	Mouse leukemic monocyte macrophage cell line
ROS	Reactive oxygen species
RNS	Reactive nitrogen species
SGC-7901	Human gastric cancer cell line
SK-MEL-2	Skin melanoma
SK-OV-3	Ovary malignant ascites
SRB	Sulforhodamine B
T-47D	High PR expression of breast cancer cell line
TBHQ	Tert-butyl hydroquinon
TBARS	Thiobarbutiric acid reactive substances
TPA	Tissue plasminogen activator
U251SP	Glioma cell line
U87	Human glioma
Vero cell	African green monkey kidney fibroblasts
WHO	World Health Organization
WISH	HeLa derivative

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